A Model of Causal Quantile Treatment Effects

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Abstract

This paper develops a model of causal quantile treatment effects in the presence of endogeneity. The causal effects of interest are defined in the potential outcomes framework. The quantile treatment effect is the difference between the quantiles of potential outcomes under the different treatment plans. This difference need not be the quantile of the difference of outcomes. Such formulation incorporates the treatment model of Doksum-Lehmann as a special case. Since the realized treatment is typically self-selected, the conventional quantile regression methods of estimating these causal effects are inconsistent. Thus, the model links treatment selection to (1) instrumental variables that do not affect the potential outcomes and to (2) the unobserved components that depend on the potential outcomes. The model requires *neither* monotonicity assumptions nor independence between the instrumental variables and the potential treatment status, both of which are at the heart of the prevalent IV models. Instead it requires an assumption called here similarity, a generalized notion of rank invariance. This assumption is reasonable in many applications.

The model is developed and illustrated with two economic problems – returns to education and demand analysis. In the first case, the quantile treatment effects help determine the effects of education on the quantiles of potential earnings. In the second case, the quantile treatment effect is an elasticity of the demand function that depends on the level of the demand, generalizing conventional models where the demand elasticity is independent of the demand level.

Identification of the model does not require functional form assumptions. Identification of non-parametric and semi-parametric forms is discussed. The identification conditions as well as estimation procedures derived from it, whether parametric or non-parametric, are different from, yet related to, the conventional quantile regression. In fact, we characterize all consistent estimation procedures as *inverse quantile regressions*. The proposed model and the inverse quantile regression estimators are useful in a wide variety of economic applications.

[1]©2001, Victor Chernozhukov and Christian Hansen. This is an incomplete and provisional version. Updates can be obtained at `vchern@mit.edu`. Please do not circulate this paper. We thank the seminar participants at U. Penn for comments, especially Y. Kitamura, P. Todd, and F. Diebold.

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1 Introduction

Quantile regression models, Koenker and Bassett (1978), describe the quantiles of a variable and their dependence on other explanatory factors. The ability of quantile regression models to capture this wealth of effects has motivated their development and led to numerous empirical applications in both economics and statistics. Aspects of quantile modeling and estimation were developed in the fundamental works of Koenker and Bassett (1974, 1978, 1982); Doksum (1974), Powell (1986); Manski (1975, 1985); Horowitz; Portnoy (1991); Chamberlain (1992); Buchinsky (1994); Buchinsky and Hahn (1996); Abadie, Angrist, and Imbens (2000); and others. Koenker and Hallock (2001) also offer a review of applications of quantile regression.

The ability of quantile regression models to characterize the impacts of variables on the distribution of outcomes and to capture diverse distributional effects makes them appealing for examining many economic applications. The distributional impacts of social programs, such as welfare and unemployment insurance, are of primary interest to policy makers. The effect of education on earnings is also a topic which has spawned numerous studies in labor economics in which the distributional impacts are important. Unfortunately, in all of these situations, treatment is self-selected or endogenous, rendering conventional quantile regression invalid.

In this paper, we propose a model of quantile treatment effects (QTE) which allows for endogeneity of treatment status. We develop the model in the potential outcomes framework and define the QTE as the difference in the quantiles of the potential outcomes under different treatment plans. This framework allows us to capture very general treatment effects and, except for the case of a simple location-shift model, is more informative than standard instrumental variables (IV) procedures. By making use of instrumental variables, like other IV methods, which affect the treatment status but not the potential outcomes, the model allows to identify the QTE. However, identification in our model does not require the independence and monotonicity assumptions that are at the heart of the prevalent IV models with heterogeneous treatment effect (the latent index or LATE models). Instead our model requires an assumption called here similarity, a generalized notion of rank invariance. This assumption is reasonable in many applications.

The remainder of the paper is organized as follows. Section 2 contains a brief outline of the previous work. Section 3 presents our model. In Section 4, we illustrate the model through two examples: demand analysis and the returns to schooling. Section 5 presents identification results and the inverse quantile regression. Estimation methods are described in Section 6, and Section 7 contains results from two empirical applications, which accompany the theoretical analysis.

2 On various proposals

This section briefly summarizes various efforts to develop models for causal quantile treatment effects with non-experimental data. It stresses the difficulty of formulating such models in the presence of endogeneity, thus posing this problem as the main research question, leaving estimation questions aside.

2.1 Model of Amemiya

The model consists of two equations

$$\begin{aligned} &(i) \quad Y = D'\delta + X'\beta + U, \\ &(ii) \quad D = Z'\gamma + V, \end{aligned} \tag{1}$$

where D is an endogenous vector, i.e. D depends on the real-valued U, X is a vector of exogenous or predetermined variables, U and V are independent of X and Z, and U and V are jointly symmetric and absolutely continuous.

Parameters (β, δ) can be estimated by two-stage LAD (Amemiya (1982)), by projecting D on Z to get $\hat{\gamma}$, and using the median regression of Y on $(Z'\hat{\gamma}, X)$. An alternate second stage is the quantile regression (Chen and Portnoy, 1993). Model (1) imposes the *constancy* of TE. The treatment variable, if assigned externally, shifts the location of the outcome variable, but does not affect the scale or shape of its distribution. This limits the treatment variety to the shift effects.

The assumptions of a constancy QTE is crucial for validity of the 2SQR procedure of Chen and Portnoy. When the QTE effects are non-constant, the 2SQR neither consistently estimates δ nor the QTE.

Unfortunately, a rather extensive empirical literature uses 2SQR to estimate the *non-constant* QTE. Such a confusion underscores the difficulty of formulating QTE models with non-constant QTE. In such cases, we must admit a form of dependence between D and U which is not related to "endogeneity," and it suddenly becomes hard to speak of "endogeneity" or "exogenity" in such terms. Once the treatment is permitted to affect other features of the outcome variable than its location, such form of dependence must be present even when D is externally assigned.

2.2 Abadie-Angrist-Imbens Model.

In Abadie, Angrsit, Imbens(2000), henceforth AAI, the treatment variable D and the instrument Z are both *binary*. The binary nature of these variables is critical, and unfortunately the extensions to the general case are not known at this point. This approach, however, is well suited to many experimental studies with imperfect compliance.

The potential outcomes Y_d are indexed by the treatment status $d \in \{0, 1\}$, and the potential treatments D_z are indexed by the instrument status $z \in \{0, 1\}$. The

realized outcome is $Y = Y_D$, while the realized treatment is $D \equiv D_Z$. AAI impose the independence condition

$$(Y_0, Y_1, D_1, D_0) \text{ are independent of} Z, \tag{2}$$

and the monotonicity assumption:

$$D_1 \geq D_0 \text{ a.s.} \tag{3}$$

For example, let

$$D_Z = 1(\varphi(Z) > V), \qquad Z \text{ is independent of } V, \tag{4}$$

i.e. $D_0 = 1(\varphi(0) + V)$ and $D_1 = 1(\varphi(1) + V)$. V may depend on the potential outcomes Y_0 and Y_1. Model (4) along with (2) satisfies the independence and monotonicity assumption. (Vytlacil (2000) also shows that the converse is true as well in the sense of distributional equivalence.)

Using the binarity of D and Z independence, and monotonicity, AAI show that in the *subpopulation of compliers*, where $D_1 > D_0$, the realized treatment D is independent of potential outcomes:

$$(Y_1, Y_0) \text{ are independent of } D \mid X, D_1 > D_0.$$

This suggests a way to study the causal effects in this sub-population. The compliers are manipulated by the instrument and, therefore, randomly receive a treatment status. That is, the treatment status is given to them independently of their potential responses Y_0 and Y_1, conditional on observed covariates X.

Let $Q_{Y|X,C}(\tau)$ denote the τ-quantile for the population of compliers conditional on $(X, D_1 > D_0)$. The quantile treatment effect $\delta(\tau)$ is a difference in the conditional τ-quantiles of Y_1 and Y_0 for compliers:

$$Q_{Y|X,C}(\tau) = \delta(\tau)D + X'\beta(\tau).$$

AAI suggest an *ingenious* weighting scheme that "finds compliers" and interpret their estimator as a re-weighted Koenker and Bassett's quantile regression.

2.3 Wald's IV approach.

The following is a Wald type IV quantile restriction

$$P(Y \leq X'\beta | Z) = \tau, \tag{5}$$

for some random vectors X and Z. The "idea" is to form a threshold function of X which Y exceeds with probability $1 - \tau$, conditional on "instruments" Z.

Such IV restrictions have been suggested by numerous people (mostly conversations), but none have linked it to the causal inference. This equation has lacked a

credible model behind (beyond the statements that the equation "allows endogeneity"). Indeed, *how should we interpret* β? For example, within the conventional IV models (e.g. latent index or LATE models as in AAI), the "parameter" β generally has no meaning.

In statistics, a form of (5) was first used by Hogg (1975), who used Wald's instruments in an attempt to formulate an interesting (graphical) quantile estimator in a regression model. See also Heggerty and Pepe (1999). In econometrics, these restrictions have been considered by Christophersen et. al. (1998), MaCurdy (1999), Hong and Tamer(1999), among others. Following Koenker (1998), we call (5) the Wald IV quantile restriction.

3 A Model of Quantile Treatment Effects

Section 3.1 defines QTE as a causal estimand via a potential outcomes framework. Section 3.2 briefly revisits a motivating model of treatment effect by Lehmann and Doksum, which Doksum axiomatically linked to the QTE. Section 3.3 presents the QTE model of this paper.

3.1 Potential Outcomes and the QTE

Neymann and Fisher were the early proponents of the causal inference and its experimental interpretation. The "Logic of the Lab" (Fisher), that is, an (possibly hypothetical) experiment (Neymann) or a theoretical model define the causal estimands of interest.

Suppose the treatment is $d \in \mathcal{D}$, where $\mathcal{D}$ is a subset of $\mathbb{R}^l$. For example, d is a crop variety in Fisher's field experimentation example, an education level in the returns-to-education analysis, or a price level in demand function models. Of interest are the random real-valued *potential outcomes*

$$\mathcal{Y}(\omega) = \{Y_d(\omega), d \in \mathcal{D}\},$$

indexed by the treatments d. The outcomes are assumed to be defined for each $d \in \mathcal{D}$ and state of the world $\omega \in \Omega$. In the social program evaluation, ω often represents an individual from a population of interest. In Fisher's agricultural example, ω is a particular plot. In the demand example, ω is a state of the world.

The *treatment effects*

$$Y_d(\omega) - Y_d'(\omega)$$

are generally *nonconstant*, they vary across ω. This is known as the *treatment heterogeneity*. Treatment heterogeneity is very important in modeling the impact of social programs and serves as an important motivation for the natural experiments literature.

Given the actual treatment D, the observed outcome is

$$Y \equiv Y_D.$$

Thus generally only the D-th component of $\mathcal{Y}$ is observed, so the treatment effect is not identified from the distribution of the realized outcomes and treatments (Y, D).

Thus the causal effects of interest, and derived from it welfare analysis, are formulated in terms of marginal distributions of outcome variables under the different treatment plans. For example, $\mu_{d,d'} = EY_d - EY_{d'}$ is the mean treatment effect (MTE). The QTE is the difference in the quantile functions[2]

$$\Delta_{d,d'}(\tau) = Q_{Y_d}(\tau) - Q_{Y_{d'}}(\tau).$$

3.2 Doksum's Treatment Function and Proneness.

Doksum(1974), Lehman(1974), Koenker and Geling(2000) present an interesting framework for describing treatment effects. Lehman (1974, Section 2.2) begins with the following model of how a treatment may operate:

> "Suppose that there exists a function $\delta(\cdot)$ such that the treatment adds the amount $\delta(y)$ when the response of the untreated subject is y. Then the distribution of the treatment response V is that of the random variable $Y + \delta(Y)$."

Doksum (1974) postulates two axioms:

(i) $\delta(\cdot) = 0$ if and only if the treatment distribution is the same as the untreated distribution, and

(ii) an equivariance of $y + \delta(y)$ to non-decreasing transformations of y,

which implies $y+\delta(y)$ is non-decreasing and thus, supposing the untreated distribution is F_{Y_o} and the treated distribution is F_{Y_1},

$$F_{Y_1}(y) = F_{Y_o}(y + \delta(y)), \text{ or } \delta(y) = F^{\leftarrow}_{Y_1}(F_{Y_o}(y)) - y.$$

Doksum(1974) calls $\delta(y)$ the *treatment effect function*. Evaluating $\delta(y)$ at the quantiles of the untreated distribution $y = Q_{Y_o}(\tau) = F^{\leftarrow}_{Y_o}(\tau)$, we get the QTE

$$Q_{Y_1}(\tau) - Q_{Y_o}(\tau).$$

This interpretation has been developed by Koenker.

In this model, subjects with the different control response will react differently to the treatment. Doksum notes that the subjects "can be said to have a certain property such as 'prone to die at an early age', 'prone to grow tall', 'prone to get high blood pressure', 'prone to learn fast', etc." The treatment function measures the *interaction* between the proneness and the treatment. Doskum argues that such a model is a much

[2]Generally, QTE are more informative than MTE, since they summarize the distributional impact, whereas MTE summarize the impact on the first moment of the distribution. In fact,

$$\mu_{d,d'} = \int_0^1 \Delta_{d,d'}(\tau) d\tau.$$

better approach to real problems than the model of a constant treatment effect which assumes that *all subjects* react in the *same way* to the treatment regardless of their control response.

Doksum also points that the model "assumes that the *two subjects* with the *same control* response will react *in the same way* to the treatment." Thus subjects react differently to the treatment if and only if they differ in this latent characteristic. We call this property *rank invariance*.

The DL model can be rewritten as follows. For some treatment o,

$$\begin{aligned} &\{Y_d(\omega) = Y_o(\omega) + \delta_d(Y_o(\omega)), \quad \forall d \in \mathcal{D}, \quad \forall \omega \in \Omega\}, \\ &\delta_d(y) = F^{\leftarrow}_{Y_d}(F_{Y_o}(y)) - y. \end{aligned} \tag{6}$$

The model is general, with the main restriction being rank invariance.

Thus Doksum, Lehman, and Koenker produced a very interesting inference paradigm. While not adopting its formal side, we develop a model that (1) incorporates treatment endogeneity and (2) retains the informal side of this paradigm, by allowing similar interpretations to apply, while permitting substantive non-invariance. (1) is the main contribution of this work, while contribution (2) facilitates the interpretation of the data-analytic findings.

3.3 The Quantile Treatment Model with Endogenous Treatment.

This model consists of two parts: one posits the potential outcomes model. The other part relates the treatment choice to the potential outcomes.

Each *potential outcome* $Y_{d,X}$ is indexed by the *treatment* d and observed *characteristics* X. The indexing serves to denote the random variables that are different, e.g. $Y_{d,x}$ is a different random variable than $Y_{d,x'}$. The indexing by X is suppressed.

All variables are defined on the probability space $(\Omega, \mathcal{F}, P)$. Measurability of all functions and variables is assumed, unless otherwise stated.

Assumption 1 (Principal Assumptions of QTE Model) Variables (X, Z) take values in $\mathcal{C}$, a subset of $\mathbb{R}^{p+k}$, d is an element of the set $\mathcal{D}$, a subset of $\mathbb{R}^l$. $\mathcal{D}$ contains at least one element, denoted o. For P-almost every value of (X, Z),

A1 Potential Outcomes. For a non-decreasing, left-continuous function $u \mapsto q(d, x, u)$, mapping $(0, 1)$ to $\mathbb{R}$, given $X = x$ and r.v. $U_d \stackrel{d}{=} U(0, 1)$,

$$Y_d \equiv q(d, x, U_d)$$

A2 Selection. For an unknown map $\delta(z, x, \cdot)$ and random process V_z

$$D_z \equiv \delta(z, x, V_z).$$

A3 Independence. Given X,

$$\{U_d\} \text{ is independent of } Z.$$

A4 Similarity. For each d, given (V, X, Z)

$$U_d \text{ is equal in distribution to } U_o.$$

A5 Observables consist at least of

$$\begin{cases} Y \equiv q(D, X, U), \\ D \equiv \delta(Z, X, V), \\ X, Z. \end{cases}$$

where $U \equiv U_D, V \equiv V_Z$.

Remark 3.1 In the preceeding assumption $\{Y_d\}, D_z, \{U_d\}$, V_z are indexed by the observed covariate values X. That is, $Y_d \equiv Y_{d,X}$, $U_d \equiv U_{d,X}$, $V_z \equiv V_{z,X}$. This dependence is suppressed as is conventional in the literature.

Remark 3.2 The spirit of our model, except ofcourse A1, is largerly motivated by the work of Vytlacil (2001), albeit we need neither the monotonicity assumptions nor the independence between V and Z (independence of D_z and Z). We only need A3.

The following briefly explains the model, while subsequent sections elaborate.

1. Causal Effects. The model has the following treatment effect interpretation. The estimand of interest is the **Conditional Quantile Treatment (Exogenous) Effect**

$$\Delta(\tau, x) \equiv q(d, x, \tau) - q(o, x, \tau),$$

that is, the difference in quantiles of potential outcomes distributions, conditional on the observed characteristic x. If d is externally (exogenously) assigned, the difference in quantiles of outcomes is $\Delta(\tau, x)$.

Indeed, (A1) defines the *potential outcome* variables $Y_{d,x}$ when treatment d is set externally to d, for the subjects with observed characteristic $X = x$. $Y_{d,x}$ has the τ-quantile at

$$Q_{Y_{d,x}}(\tau) = q(d, x, \tau).$$

Thus, assumption (A1) is always feasible, since conditional on $X = x$, $Y_d \overset{\text{a.s.}}{=} Q_{Y_{d,x}}(U)$, for some uniform r.v. U. All we require is that Y_d be defined.

2. Treatment Heterogeneity. (A1) embodies very general forms of heteroscedasticity – the treatment d and covariates x can affect location, scale, and the *entire shape*

of the distribution of potential outcome $Y_{d,x}$. This effect is realized through the mapping $(d, x, \tau) \mapsto q(d, x, \tau)$. Thus (A1) implies considerable *treatment heterogenity* in that the *treatment effects vary* across the states $\omega \in \Omega$, $\mathrm{TE} = Y_{d,x} - Y_{d',x} \neq$ const.

3. Ranking. If we are to speak of ω as indexing individuals, then, for the same observed characteristic x and treatment d the subjects *differ*. Their ranking with respect to each other is determined by the random variable $U_{d,x}$. We call vector $(U_{d,x}, d \in \mathcal{D})$ the *rank* or *ability* vector. Rank invariance, postulated by Doksum, is an important special case:

$$U_{d,x} \equiv U_{d',x}, \text{ for each } d, d'. \tag{7}$$

4. Selection. (A2) describes how selection of treatment is made. $D_{z,x}$ is the random variable which represents the choice of the potential treatment status made by individuals with characteristic x in the case when the instrument status is z. The random vector (which could be infinite-dimensional) $V_{x,z}$ is responsible for the difference in decisions across observationally identical individuals. $\delta(\cdot)$ is the *selection function* that maps the tuple (x, z, v) into the *treatment set* $\mathcal{D}$. This function is unknown and is not assumed to be monotone in *the process* V.[3]

5. Instrument Independence. (A1) assumes an *exclusion restriction* – the potential outcomes are not shifted by the instrument variable Z. (A3) states that potential outcomes $\{Y_{d,x}\}$ (determined by $U_{x,z}$) are *jointly independent* of the instrumental variable Z. This assumption is much more general than the assumption, which is conventional in the literature, that states that not only the potential outcome but *also the potential treatment status* $\{D_z\}$ *are independent of the instrument* Z. This is a very strong assumption that is easily violated when the instrument is measured with error or there are omitted variables related to Z (a part of error V) in the selection equation.

6. Similarity. (A4) is weaker than the invariant rank (common ability) assumption (7), yet it is not most general assumption. More generally, the treatment decision may incorporate *ex-post* knowledge of rank vector $U_{d,x}$ across all d. (A4) states that conditional on the information (V, Z, X), where V is unobserved, we should forecast the same ranks U_d across the treatment states d. In other words, *ex-ante* the ranks are "similar". This assumption is more restrictive than the conventional IV approaches. However, (A4) enables a much more general decision function in (A2) than the monotonicity assumption or independence restriction in equation (2). Thus, when confined to the binary case, our model is neither more nor less general than AAI model.

7. Observables. (A5) states what is observed in the process of data-generation.

[3] A recent model of Imbens and Newey (2001) uses a selection function that is one-to-one in the error variable V and heavily exploits that, along with independence between V and Z, to identify their model. No such strong assumption is needed here.

4 Nature of the Model

The section considers examples and draws some general comparisons.

4.1 Examples

The following section embeds a generalization of the classical demand-supply example. This example illustrates that many classical simultaneous equation models can be written as special cases. Furthermore, what is presented is perhaps one of the most general simultaneous equation models that one can find in the literature. Another example concerns the causal effects of education.

Example 4.1 (Demand Example) Consider the following model

$$\begin{cases} \textbf{i.} & Y_p = q\left(p, U\right), \\ \textbf{ii.} & \tilde{Y}_p = \rho\left(p, z, \mathcal{U}\right), \\ \textbf{iii.} & P \ \in \{p : q\left(p, Z, U\right) = \rho\left(p, \mathcal{U}\right)\} \end{cases} \tag{8}$$

The map $p \mapsto Y_p$ is the random demand function, that is the *potential demand* when the price is set (externally) to the value p. Likewise, $p \mapsto \tilde{Y}_p$ is the random supply function, that is the *potential supply* when the price is set (externally) to p. Additionally, Y_p and $\tilde{Y}_p$, $q(\cdot)$, and $\rho(\cdot)$ depend on the covariates X, but this dependence is suppressed. Random variable U is the level of the demand in the sense that

$$q(p, U) \leq q(p, U') \text{ when } U \leq U'.$$

Demand is maximal when $U = 1$ and minimal when $U = 0$, holding p fixed. Likewise, $\mathcal{U}$ is the level of supply. The τ-quantile of the demand curve $p \mapsto Y_p$ is given by

$$p \mapsto Q_{Y_p}(\tau) \equiv q(p, \tau).$$

Thus with probability τ, the curve $p \mapsto Y_p$ lies below the curve $p \mapsto Q_{Y_p}(\tau)$.

The quantile treatment effect is characterized by an elasticity

$$q(p', \tau) - q(p, \tau) \quad \text{ or, if defined, } \quad \partial \ln q(p, \tau) / \partial \ln p.$$

The elasticity depends on the state of the demand τ (low or high) and may vary considerably with τ. For example, this variation could arise when the number of buyers varies and aggregation induces non-constant elasticity across the demand levels, as a process of summation of individual demand curves, holding the price fixed.

This model incorporates many traditional models of the following sort

$$Y_p = q(p) + \mathcal{E}, \text{ where } \mathcal{E} = Q_{\varepsilon}(U). \tag{9}$$

The model (8.**i**.) is much more general in that the price can affects the distribution of the demand curve, while here it only affects the location of the distribution of the demand curve. This translates into *non-parallel fluctuations* of the demand curve.

Condition **iii**. is the equilibrium condition that generates *endogeneity* – the *selection* of the actual price by the market depends on the potential demand and supply outcomes **i.** and **ii.** As a result

$$P = \delta(Z, V),$$

where V consists of U, $\mathcal{U}$, and other variables ("sunspot" variables, if equilibrium price is not unique.) Thus what we observe can be written as simultaneous equations of a very general form (A5)

$$\begin{cases} Y \equiv q(P, U), \\ P \equiv \delta(Z, V). \end{cases} \tag{10}$$

Because of endogeneity, it is generally true that $Q_{Y|P}(\tau) \neq q(P, \tau)$, therefore the conventional quantile regression will be an invalid procedure to estimate the τ-th quantile demand curve. Additionally, it can be shown that the two-stage quantile regression, described in section 2.1 is generally not suitable for estimation purposes.

We show that the presence of instrumental variables Z, like weather conditions, that shift the supply curve and do not affect the level of the demand curve U allows to identify the $\tau - th$ quantile of the demand function, $p \mapsto q(p, \tau)$. Furthermore, our model allows arbitrary correlation between Z and V to exist. This allows, for example, measurement error in Z (e.g. in weather conditions). The standard IV approaches (Heckman et. al.(2001), Imbens and Angrist (1994), Vytlacil) do not seem to accommodate such a possibility.[4]

Example 4.2 (Schooling Returns) Let "earnings" in the schooling states $d \in \{0, 1\}$ be determined by a general "random coefficients" model

$$\log(Y_{1,X}) = q_1(X, U_{1,X}),$$
$$\log(Y_{0,X}) = q_0(X, U_{0,X}).$$

An individual's decision is given by

$$\begin{cases} 1 \text{ if } \varphi(Z, X, V) \geq 0, \\ 0 \text{ if } \varphi(Z, X, V) < 0, \end{cases}$$

where $V \equiv V_{Z,X}$ potentially depends on the ability vector $(U_{1,X}, U_{0,X})$ and arbitrarily on functions q_1 and q_0, X and Z. The first kind of dependence is a cause of endogeneity.

In the standard Roy model, no restrictions are placed on the individual specific variations in earnings, and it is generally assumed that the individual observes these before making the schooling choice. In order to identify the quantile treatment effect, we assume that, conditional on (Z, X, V), $U_{1,X} \stackrel{d}{=} U_{0,X}$. This is more restrictive than

[4] Of course, for this to matter empirically, the violation of the independence condition has to be sizable. Thus the plausibility of one or the other model is a matter of approximation. Heckman et. al (2001) provide ingenious tests that enable to test such a violation.

the general case, but perhaps not as restrictive as it may appear at a first glance. This restriction allows arbitrary correlation patterns to exist between the two distributions, and allows the treatments to have different impacts in the two states through the $q_1(\cdot)$ and $q_0(\cdot)$ functions. One chief difference between the model presented here and the standard Roy model is the implicit ex ante nature of the decision process. Instead of knowing the exact outcomes in any state of the world, the decision maker anticipates the same distribution of ability across treatment states and makes the decision accordingly.[5]

Finally note that the equality in distribution only need hold conditional on Z, X, and V_Z. Overall, this seems to be a reasonable framework for examining the returns to schooling. For example, people generally decide on whether to attend college or not before they observe exactly where they lie in the ability distribution of college educated and non-college educated individuals and are likely to anticipate the same distribution of ability **relative to similar** individuals (with the same covariate X) across the treatment states.

Another main difference with Roy models, as used in econometric modeling, is that our approach expressly allows for dependence to exist between the instrument Z and V whereas the standard approaches expressly disallow this, as mentioned in the previous example. E.g., consider the following simple schooling decision rule

$$D = 1\{\varphi(Z) + V_Z \geq 0\}.$$

In the schooling context, if Z is a cost of education or family background, it may be measured with a sizable error, so independence between Z and V need not hold. V_Z could also capture omitted variables which are correlated to Z and impact the schooling decision but not the outcome. The same causes also violate the monotonicity assumption often used in the IV literature.[6]

4.2 Remarks after Examples

Three aspects of the proposed model are highlighted by the stated examples. The first is the the richness of treatment effects embedded in the model. The second is the usefulness of the similarity assumption in identification as well as interpretation of the quantile treatment effects. The third is the rich structure of the selection equation.

(a) The Structure of Quantile Treatment Effects.

The proposed model allows *arbitrarily general* quantile treatment effects. The treatment can affect the entire shape of the outcomes distribution, inducing rich forms of treatment heterogeneity. The effect is realized through the mapping

$$(\tau, d, x) \mapsto q(d, x, \tau)$$

[5] More precisely, we assume that he has enough information only to anticipate the same distribution of ability across states. The assumption does not require the decision-maker to have correct beliefs.

[6] See Imbens and Angrist (1994) for other examples of violation.

that maps quantile index τ, the treatment status d and observed controls x into the τ-quantile of the potential outcome variable $Y_{d,x}$. $Y_{d,x}$ is the outcome when the treatment is (externally) assigned to be d and observed characteristics equal x. $Y_{d,x}$ varies among individuals or states of the world ω, and the quantile function $\tau \mapsto q(d, x, \tau)$ is a complete description, a model of this variation. Note that mean functions or any other simple summaries are generally insufficient to describe the outcome distribution.

For a given τ, the *τ-quantile treatment function*

$$d \mapsto q(d, x, \tau)$$

describes the τ-quantile of potential outcome as a function of the treatment. In the schooling example, $d \mapsto q(d, x, \tau)$ traces dependence of earnings on the *exogenously determined* schooling level, that is, when the schooling level is externally set. In the demand analysis example, the QT function $p \mapsto q(p, \tau)$ describes the τ-th quantile function of the demand curve. $q(p, \tau)$ is indexed by τ which denotes the demand level, with higher τ corresponding to higher demand levels. Thus quantile treatment effects describe the variation as well as distribution of the elasticity across the demand levels.

(b) Similarity Assumption and Doksum's Proneness

When interpreting the nature of the assumed treatment effect, it is useful to explicitly discuss the rank or ability vector $(U_{d,x}, d \in \mathcal{D})$. Ex-ante, the ranks across the treatment states are all similar in that the forecast of (any of function of) the rank, given the information (X, Z, V) is the same, while the ex-post, the ranks may differ. Thus from the ex-ante perspective the quantile index τ is an ex-ante rank, ability, or Doksmus's proneness. Thus, in the spirit of the Doksum-Lehmann-Koenker model, the quantile treatment effect $\Delta(x, \tau)$ measures the interaction between this unobserved latent ex-ante ability and the treatment. The similarity assumption seems to be a plausible approximation to many settings, as discussed in the previous section. In the demand example, rank invariance, a special case of the similarity assumption, is, in essence, a condition for continuity of the demand function (if we are willing to assume it). See Doksum (1974), Robbins, Vytlacil (2001) for other justifications of rank invariance.

(c) Structure of Selection Equation

The selection equation in (A2),(A3) is very general, and in fact, is considerably more general that those in the popular IV approaches, albeit at the cost of the similarity assumption. (A2) -(A3) require *neither* independence between the potential treatment status and the instrument *nor* monotonicity assumptions. This is important, since errors occurred in the measurement of Z or omission of variables related to instruments in the selection equation generally lead to the violation of either of these assumptions. On the other hand, the approach does requires the similarity assumption, which is a generalized notion of rank invariance. Overall, while the gain in flexibility its value must be weighted against the similarity assumption in each application.

5 Identification and Inverse Quantile Regression

Here we establish a link between the QTE model A1-A5 with the Wald's IV restrictions and quantile restriction. We also show that the treatment effects are identified without functional forms assumptions. Identification without relying on functional forms is crucial to any sensible treatment effect model. While good functional forms can be used to sharpen the estimation, the nonparametric identifiability is generally crucial to a universal application of any treatment effect model.

5.1 Main Identification Restriction

The following theorem provides provides a fundamental link of the parameters of model A1-A5 to the Wald's IV estimating equations.

Theorem 1 (A1-A5 meet Wald) *Suppose A1-A5 hold, and*
i. *$q(D, X, \tau)$ is strictly increasing in τ, P-a.e., then*

$$\begin{aligned} P\left[Y \leq q(D, X, \tau) | X, Z\right] &= \tau, \\ P\left[Y < q(D, X, \tau) | X, Z\right] &= \tau, \end{aligned} \tag{11}$$

ii. *$q(D, X, \tau)$ is non-decreasing in τ, P-a.e., then*

$$\begin{aligned} P\left[Y \leq q(D, X, \tau) | X, Z\right] &\geq \tau, \\ P\left[Y < q(D, X, \tau) | X, Z\right] &\leq \tau. \end{aligned} \tag{12}$$

Remark 5.1 (Discrete vs. Continuous Outcomes) Case **i.** corresponds to the continuous potential outcomes Y_d. Case **ii.** corresponds to non-continuous (binary, discrete, or mixed) potential outcomes, and so is of relevance to studying the quantile treatment effects in such settings. See Manski (1985) for the discrete cases when there is no endogeneity. See also Kordas(2000).

Theorem 1 links the causal quantile treatment effect model A1-A5 and the Wald's IV quantile restrictions. Theorem 1 thus provides an empirical and causal content to these restrictions. In this regard, the QTE model A1-A5 serves the same purpose as the LATE model developed by Imbens and Angrist (2000) to provide the link between the Wald's IV approach and mean causal treatment effects. Our results concern with the causal quantile treatment effects and employ a different modeling philosophy to obtain the link.

There is a fine road leading to Theorem 1, as Examples 5.1 and 5.2 below demonstrate. In addition, it is worth stressing that *neither* independence between the potential treatment status and the instrument *nor* monotonicity assumptions, both at the heart of popular IV approaches, are needed for Theorem 1 to hold. On the other hand, the approach requires the similarity assumption, which is a generalized notion of rank invariance. See the earlier discussion.

As noted, Theorem 1 allows for arbitrary (discrete or continuous) variable Y, for arbitrary treatment variable D, or instrument Z. Thus equations (11) and (12) lead to natural ways to estimate any model with endogeneity as long as the corresponding quantiles of the potential outcome distribution

$$q(d, x, \tau)$$

may be specified. One, for example, can form the estimating equations based on (11) or (12) and proceed with the estimation. Note that typically it is not possible to use any other constraint of the form

$$E\left[\psi(Y, X, g)|Z\right] = 0,$$

except of the form in (11) and (12), as shown below.

Example 5.1 (Nonorthgonality of Mean Treatment Function) Within A1-A5, suppose $\mu(d)$ is finite in the equation

$$Y_d = \mu(d) + \epsilon_d, \quad E\epsilon_d = 0;$$

$\mu(d)$ is the mean treatment function. It would be natural to expect that

$$E\left[Y - \mu(D)|Z\right] = 0,$$

but this is **false** since generally

$$E\left[q(D, U) - \mu(D)|Z\right] \neq 0,$$

because U is not independent of D conditional on Z in general, so that

$$\begin{aligned} E\left[q(D, U)|Z\right] &= \int_{\mathcal{D}} \int_{[0,1]} q(d, u) dP[D = d, U = u|Z] \\ &\neq \int_{\mathcal{D}} \int_{[0,1]} q(d, u) dP[D = d|Z] \cdot dP[U = u|Z] \equiv E\left[\mu(D)|Z\right]. \end{aligned}$$

The equality holds when there is no endogeneity or when the treatment effect is constant.

Example 5.2 (Non-orthogonality of Probability Treatment Function) Suppose we are interested in the probability treatment function

$$P\left[Y_d \leq t\right] = p(d, t).$$

Then again it may be intuitive to expect in view of Theorem 1 that

$$E\left[1\left[Y \leq t\right] - p[D, t]|Z\right] = 0.$$

Again, because U is not independent of D conditional on Z, this is **false**, since

$$\begin{aligned} P\left[Y \leq t|Z\right] &= \int_{\mathcal{D}} \int_{[0,1]} 1\left[u \leq p(d,t)\right] dP[D=d, U=u|Z] \\ &\neq \int_{\mathcal{D}} \int_{[0,1]} 1\left[u \leq p(d,t)\right] dP[D=d|Z] dP[U=u|Z] \equiv E\left[p(D,t)|Z\right]. \end{aligned}$$

Note again that if they are independent, then there is no endogeneity.

5.2 The Inverse Quantile Regression

The main identification restriction of Theorem 1 can be posed as an optimization problem, which we call the *inverse quantile regression* for its "inverse" relation to the (conventional) quantile regression, Koenker and Bassett (1978). This links our model, the Wald IV restrictions, and quantile regression together.

In order to do that, we note that Theorem 1 essentially states that 0 **is** the τ-th quantile of random variable $Y - q(x,d,\tau)$ conditional on $(X = x, Z = z)$. Therefore, the problem of finding a function $q(x,d,\tau)$ satisfying equations (11) and (12) is the problem of the *inverse quantile regression*: find a function $q(x,d,\tau)$ such that **0** is the solution to the quantile regression problem, where we regress $Y - q(x,d,\tau)$ on any function of (z,x), that is, find $q(d,x,\tau)$ such that

$$Q_{Y-q(D,X,\tau)|X,Z}(\tau) = 0 \quad P\text{-a.e.}$$

Theorem 2 is formal statement of this result. For each $X = x$ consider the set $\mathcal{S}(x)$ of measurable functions φ mapping from the range $(D|x)$ into the range $(Y|x)$. Clearly, $\mathcal{S}(x)$ is non-empty, since $d \mapsto q(d,x,\tau)$ is one such function.

Theorem 2 (Inverse QR: Wald meets Koenker and Bassett) *For P-a.e. value x of X, the following are **equivalent** statements,*

1. *φ satisfies equation (10)-(11).*
2. *$Q_{Y-\varphi(D,X,\tau)|X,Z}(\tau) = 0$.*
3. *φ satisfies, for P- a.e. z*

$$0 \equiv \underset{v\in\mathbb{R}}{\text{argmin}} E\left[\rho_{\tau'}\left(Y - \varphi(x,d) - v\right)|x,z\right],$$

where $\rho_\tau(u) \equiv \tau u^+ + (1-\tau)u^- \equiv (\tau - 1(u<0))u$.

4. *φ is an $\underset{\varphi'\in\mathcal{S}(x)}{\text{argmin}} \left\|v(x,z)\right\|^2$, for P- a.e. z, where*

$$v(x,z) \equiv \underset{v\in\mathbb{R}}{\text{argmin}} E\left[\rho_{\tau'}\left(Y - \varphi'(x,d) - v\right)|x,z\right].$$

Note that statements (3) and (4) involve a quantile regression, as in Koenker and Bassett (1978).

An abstract result, Theorem 2, is useful for interpretative and constructive purposes. First, *any* consistent estimation problem tries to asymptotically solve the inverse quantile regression problem. That is, any procedure tries to find the value of a quantile treatment function (or relevant structural form) such that 0 is the best predictor of $Y - q(D, X, \tau)$ under the asymmetric loss function, conditional on X, Z. Second, Theorem 2 (part (4)) suggests a way to construct estimators (in addition to obvious method of moments strategies based on the Wald equations (11) and 12). In section 6, using the analogy principle, we construct one such estimator. The estimator has a number of practical advantages over the method of moments or empirical likelihood estimators.

Observe also that Theorem 2 applies to continuous, discrete, or mixed outcomes, so that the estimation strategies based on Theorem 2 can be applied to such data (this is clearly relevant, see, for example, Manski (1975, 1985) for the case with no endogeneity).

5.3 What is needed for identification

As most papers on treatment effects, we focus on the case with discrete treatment (endogenous) variable and continuous outcomes, since this covers overwhelming majority of applications. Continuous covariate is also briefly discussed.[7] (This section is commented out, since it was not ready, but will be discussed in the talk.)

6 Estimation

This paper's main focus is the modeling. As far as estimation is concerned, Theorem 1 requires several extensive research direction (which we hope to pursue): parametric and non-parametric estimation for continuous, binary, and discrete outcomes. It is impossible to address all of these issues in this paper. However, in the next section we provide practical, effective estimators for estimating the baseline model that covers a wide area of applications of model A1-A5.

6.1 The Model

We describe the estimation of the following model, where for a given quantile index τ, the conditional τ-quantile of the potential outcome is given by

$$Q_{Y_{d,x}}(\tau) = d'\alpha(\tau) + x'\beta(\tau), \tag{13}$$

[7] This case is to be treated in detail in the work of Chernozhukov, Hansen, Imbens, and Newey.

where d is an $l \times 1$ vector of treatment variables (possibly interacted with covariates) and x is a $k \times 1$ vector of (transformations) of covariates. This model is a specialization of A1, and is a foundation of a wide variety of fundamental theoretical and empirical analyses (see sources cited on the first page of this paper).

In program evaluation studies, d is often binary and x is a set of categorical (binary) variables as well, so the model is non-parametric. In other settings the model (13) is an approximation. The model covers a wide variety of applications, including simultaneous equation models.[8]

6.2 Estimation Methods

There are several options for estimating the proposed model (13) via the Wald restrictions. One is obvious – the method of moments, as developed in Heggerty and Pepe (1998), MaCurdy (1999). However, the involved objective functions are highly *multi*-modal with zero or nearly zero derivatives over an entire parameter space, hence such estimation is infeasible when the number of parameters exceeds three, as grid searches are needed. Most applications have many covariates, so the approach can not be applied to such settings.

Here we discuss two estimators. The first is new and constitutes one of the main contributions of this paper – the inverse quantile regression. The second is an empirical likelihood estimator. In sequel, our data consists of $W_i = (Y_i, D_i, X_i, Z_i)$, $i \leq n$, which is taken to be *i.i.d.*

(a) Inverse Quantile Regression.

Inverse quantile regression is a *finite sample analog* of the inverse quantile regression in population. We assume $\alpha(\tau)$ belongs to a compact set $\mathcal{V}_\alpha$. Other regularity conditions are compactly stated as assumptions **R1-R6** in the appendix.

Define the weighted "check" function:

$$\hat{g}(W, \alpha, \beta, \gamma) \equiv \rho_\tau(Y - D'\alpha - X'\beta - \hat{\Phi}'\gamma)\hat{V}, \quad \text{where}$$

- $\Phi \equiv \Phi(X, Z)$ is a smooth $r \times 1$ vector function of an instrument,
- $\widehat{\Phi} \equiv \widehat{\Phi}(X, Z)$ is a smooth consistent estimate, satisfying **R5**,
- $V \equiv V(X, Z)$ is a smooth weight function,
- $\hat{V} \equiv \hat{V}(X, Z)$ is a smooth consistent estimate, satisfying **R5**.

Remark 6.1 Often we may *simply* set $\Phi \equiv \Phi(X, Z) = Z$ or $V = 1$, but this will not yield most efficient estimator. See remark 6.9.

Remark 6.2 Assumption **R5** allows for a very wide variety of nonparametric and parametric estimators of V and Φ. See Remark F.2.

[8] Fully nonparametric estimation covering continuous cases is developed in a companion work.

Further let

$$Q_n(\alpha, \beta, \gamma) \equiv \frac{1}{n} \sum_{i=1}^{n} \hat{g}(W, \alpha, \beta, \gamma),$$

and

$$(\hat{\beta}(\alpha), \hat{\gamma}(\alpha)) \equiv \arg\inf_{\beta,\gamma} Q_n(\alpha, \beta, \gamma), \tag{14}$$

$$\hat{\alpha}(\tau) \in \arg\inf_{\alpha \in \mathcal{V}_\alpha} \left[\sqrt{n}\left\|\hat{\gamma}(\alpha)\right\|\right]. \tag{15}$$

Plug $\hat{\alpha}$ into (14) to obtain

$$(\hat{\beta}(\tau), \hat{\gamma}(\tau)) \equiv (\hat{\beta}(\hat{\alpha}), \hat{\gamma}(\hat{\alpha})) \tag{16}$$

Equations (14) -(16) are a finite sample *inverse quantile regression*. (14) is the quantile regression step, and (14) is the "inverse" step.

Remark 6.3 (GMM vs. IQR) The nested optimization in (14)-(16) allows to effectively reduce the dimensionality of a *potentially very difficult optimization problem* to the dimension of α. By "concentrating out" the estimates of $\beta(\alpha)$ and multiplier $\gamma(\alpha)$ through the Koenker-Bassett regression quantile step (that can be solved *faster* than OLS by interior point methods), we search for the optimal α. In GMM set-up we shall be needing the grid search over a subset of $\mathbb{R}^{(\dim(x)+\dim(\alpha))}$ (e.g. in Example 2 of next section $\dim(x) + \dim(\alpha) = 16$), so such an estimator is generally infeasible, except perhaps when $\dim(x) = 2$ or 3. The inverse quantile regression needs only a grid search over a subset of $\mathbb{R}^{\dim(\alpha)}$ (and even that may not be needed – see remark (6.4)). In most applications $\dim(\alpha) = 1$, so the problem is computationally very feasible, tractable, and fast.

Remark 6.4 (Parametric Programming) Upon a further reflection, we observe that in the quantile regression step in (14), $\hat{\beta}(\alpha), \hat{\gamma}(\alpha)$ can be solved for *all* α via parametric programming, which involves finding a solution for a particular α and then using a sensitivity analysis to quickly compute the solution of all other α. This, in principle, makes the problem as computationally difficult as ordinary least squares.

(b) Generalized Empirical Likelihood. Define for $\vartheta \equiv (\alpha, \beta), \varphi_\tau(u) = \tau - 1(u < 0)$

$$\hat{f}(W, \vartheta) \equiv \varphi_\tau(Y - D'\alpha - X'\beta)\widehat{\Psi},$$

where

- $\Psi \equiv \Psi(X, Z)$ is a smooth function of an instrument,
- $\widehat{\Psi} \equiv \widehat{\Psi}(X, Z)$ is a smooth consistent estimate of such function, satisfying **L5**.

Remark 6.5 Often we may *simply* set $\widehat{\Psi} = (X, Z)$, but this will not yield most efficient estimator. See Remark 6.9.

Define also

$$Q_n(\vartheta, \gamma) \equiv \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbf{s}[\hat{f}(W, \vartheta)'\gamma],$$

$$\hat{\gamma} \equiv \arg\inf_{\gamma} Q_n(\hat{\vartheta}, \gamma), \qquad \hat{\vartheta} \equiv \arg\sup_{\vartheta \in \mathcal{V}} [\inf_{\gamma} Q_n(\vartheta, \gamma)]. \tag{17}$$

Function $\mathbf{s}(\cdot)$ is defined equal to a strictly convex, finite, and four times differentiable function $\mathbf{s}_0$ on an open *interval* of $\mathbb{R}$ containing 0, and is equal to $+\infty$ outside such an interval. Normalize $[\partial^j \mathbf{s}(v)/\partial v^j](0) = 1$ for $j = 1, 2$.

Remark 6.6 Functions $\mathbf{s}_0(v) = -\ln(1-v), \exp(v), (1+v)^2/2$ lead to the empirical likelihood (Owen), exponential tilting (Kitamura and Stutzer (1997)), and continuous up-dating GMM estimator. The GEL was first proposed by Smith.

Remark 6.7 Just like GMM, GEL is basically infeasible in our settings when we have two or more covariates. It is, however, useful in low-dimensional settings. Alternatively, it may be used as a refinement of the IQR estimator. The latter can be used to bring the estimates to a right neighborhood, and the GEL can be used to improve over such a neighborhood. For this purpose, GEL are known to have better finite sample properties than GMM. The pivotal objective function of GEL can be used for construction of confidence intervals.

6.3 Asymptotic Properies.

The properties of the proposed estimators are very different. First, we state the properties of the GEL estimators.

Theorem 3 (GEL) *Under assumptions (13),* ***L1-L6****,*

$$\begin{matrix} \sqrt{n}(\hat{\vartheta}(\tau) - \vartheta(\tau)) \\ \sqrt{n}\hat{\gamma} \end{matrix} \xrightarrow{d} N \begin{bmatrix} (J'S^{-1}J)^{-1} & 0 \\ 0 & S^{-1}[I - J(J'S^{-1}J)^{-1}J']S^{-1} \end{bmatrix}$$

where $S = \tau(1-\tau)E\Psi\Psi'$ *and* $J = E\left[f_\epsilon(0|X, Z)\Psi[D' : X']\right]$, $\epsilon \equiv Y - D'\alpha(\tau) - X'\beta(\tau)$.

Theorem 4 (IQR) *Under assumptions (13),* ***R1-R6****,*

$$\begin{aligned} \sqrt{n}(\hat{\alpha}(\tau) - \alpha(\tau)) &\xrightarrow{d} (J_\alpha' H J_\alpha)^{-1} J_\alpha' H\, \mathcal{N}(0, S) \\ \sqrt{n}(\hat{\beta}(\tau) - \beta(\tau)) &\xrightarrow{d} \bar{J}_\beta [I - J_\alpha (J_\alpha' H J_\alpha)^{-1} J_\alpha' H]\, \mathcal{N}(0, S), \\ \sqrt{n}\hat{\gamma} &\xrightarrow{d} \bar{J}_\gamma [I - J_\alpha (J_\alpha' H J_\alpha)^{-1} J_\alpha' H]\, \mathcal{N}(0, S), \end{aligned}$$

where convergence is joint, and $\mathcal{N}(0,S)$ is normal vector with mean 0 and variance $\tau(1-\tau)E\Psi\Psi'$, where $\Psi \equiv V \cdot [X' : \Phi']'$, defined in thm 3, $J_\alpha \equiv E\left[f_\epsilon(0|X,Z)\Psi D'\right]$, $J_\theta \equiv E\left[V^{-1} \cdot f_\epsilon(0|X,Z)\Psi\Psi'\right]$, $H = \bar{J}_\gamma' \bar{J}_\gamma$, $[\bar{J}_\beta : \bar{J}_\gamma]' \equiv J_\theta^{-1}$, where $\bar{J}_\beta$ is $k \times (k+r)$ matrix and $\bar{J}_\gamma$ is is $r \times (k+r)$ matrix.

Remark 6.8 A simplification occurs when $r = l$, when the number of instruments Φ is the same as the number of endogenous variables. In this case

$$
\begin{aligned}
&(J_\alpha' H J_\alpha)^{-1} J_\alpha' H = -(J_\alpha' \bar{J}_\gamma)^{-1} \bar{J}_\gamma', \\
&\bar{J}_\beta[I - J_\alpha (J_\alpha' H J_\alpha)^{-1} J_\alpha' H] = \bar{J}_\beta[I - J_\alpha (J_\alpha' \bar{J}_\gamma)^{-1} \bar{J}_\gamma], \\
&\bar{J}_\gamma[I - J_\alpha (J_\alpha' H J_\alpha)^{-1} J_\alpha' H] = 0.
\end{aligned}
$$

It then can be checked that the joint asymptotic variance of $\hat{\alpha}$ and $\hat{\beta}$ equals:

$$J^{-1} S J^{-1},$$

for J defined in Theorem 3.

Remark 6.9 When we further pick $\Phi = E[D|Z,X]$, $V = f_\epsilon(0|X,Z)$ and $\Psi = V \cdot [X', \Phi']'$, variance of *both* estimators simplifies to the efficiency bound

$$\tau(1-\tau)E[\Psi\Psi']^{-1}.$$

This expression is very convenient to work with and estimate. $f_\epsilon(0|X,Z)$ can be estimated the methods described, for example, in Powell (1986) and $E[D|Z,X]$ can be estimated using series or kernel estimation (e.g. run OLS of D on Z, X and their powers). Condition **L5** allows for a great variety of such estimation procedures. It also often makes sense in moderate samples to assume $f_\epsilon(0|X,Z) \approx f_\epsilon(0)$ or approximate $f_\epsilon(0|X,Z)$ by a conditionally normal distribution.

7 Empirical Applications

7.1 Demand for Fish

In this section, we present estimates of demand elasticities which may potentially vary with the level of demand, τ. The data contain observations on price and quantity of fresh whiting sold in the Fulton fish market in New York over the five month period from December 2, 1991 to May 8, 1992. These data were used previously in Graddy (1995) to test for imperfect competition in the market and later in Angrist, Graddy, and Imbens (2000) to illustrate use of the conventional IV estimator as a weighted average of heterogeneous demands. The price and quantity data are aggregated by day, with the price measured as the average daily price for the dealer and the quantity as the total amount of fish sold that day. The data also contain information on the

day of the week of each observation and variables indicating weather conditions at sea, which are used as instruments to identify the demand equation. The total sample consists of 111 observations for the days in which the market was open over the sample period.

The top two panels of Figure 1 provides the estimates of elasticities obtained by IQR of $\ln(q_t)$ on $\ln(p_t)$ using wind speed as the instrumental variable, while the lower panels depict standard quantile regression (QR) estimates. The shaded region around the point estimates represents the 80 percent confidence interval. While the reported estimates are for a model without covariates, the estimated elasticities are not sensitive to the inclusion of dummy variables for the days or other covariates.

The price effect on quantities sold, as estimated by QR, appears to be approximately constant across the entire range of quantiles. The magnitudes of the effects are also quite small, in all cases much less than unity. IQR estimates on the other hand range from -2 to -.5, indicating a broad range of elasticities which vary with the level of demand. Except at high quantiles, the IQR elasticities are uniformly greater than the price effects predicted by QR. This is clearly shown in the demand curves plotted in Figure 2. Note that the interpretation of IQR and QR estimates is very different. IQR estimates a (causal) demand model, while QR estimates the conditional quantiles of the quantity variable as a function of the realized price variable.

The IQR estimates of the demand elasticities of the τ- quantile demand curves illustrate a considerable heterogeneity of elasticities across the demand levels. The results indicate that the elasticity of demand is quite high at low quantiles, but is decreasing in the quantile index. While there are numerous possible explanations for this demand behavior, it does cast doubt on the hypothesis that the aggregate demand in this market is made up of the sum of demand curves of numerous identical price-taking agents who randomly arrive at the market. The estimates also illustrate that a single statistic, such as the single elasticity generated by two-stage least squares, may be insufficient to truly capture the demand function variety.

7.2 Evaluation of a JTPA Program

The impact of job training programs on the earnings of participants, especially those with low income, is of great interest to both policy makers and academic economists, but evaluating the causal effect of training programs on earnings is difficult due to the self-selection of treatment status. However, data available from a randomized training experiment conducted under the Job Training Partnership Act (JTPA) provides a mechanism for addressing this issue. In the experiment, people were randomly assigned the offer of JTPA training services, but due to the fact that people were able to refuse to participate, the actual receipt of the treatment was self-selected. Of those offered treatment, only 60 percent participated in the training program. There was also a small number of individuals from the control group who received training. However, the random assignment of the training offer provides a plausible instrument

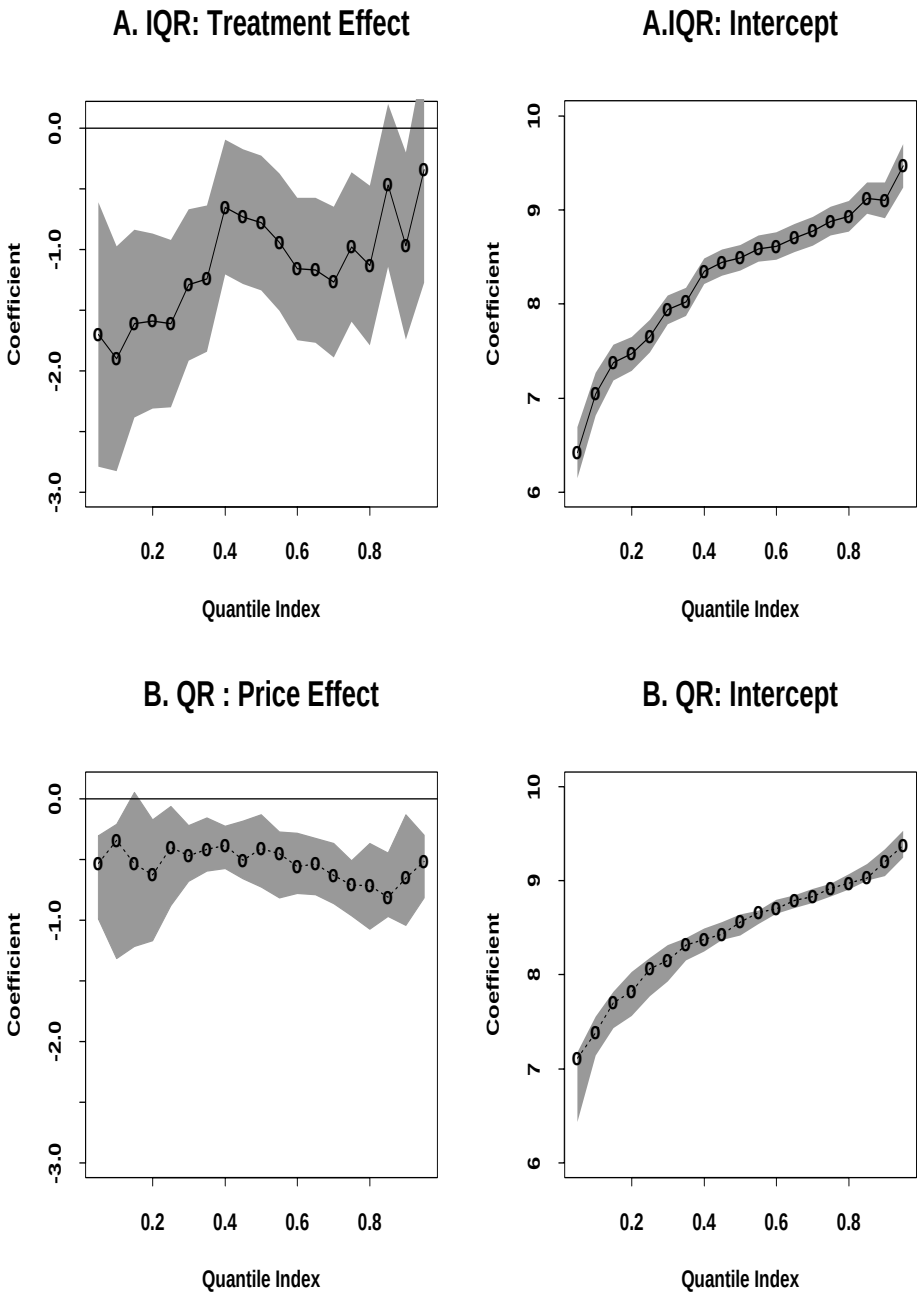

Figure 1: Inverse Quantile Regression and Quantile Regression Results for fish data. The quantile treatment effect, estimated by IQR, is the elasticity of the τ-th quantile demand curve. It tends to be much higher than the "price effect" on the τ-quantiles of quantities sold, estimated by QR.

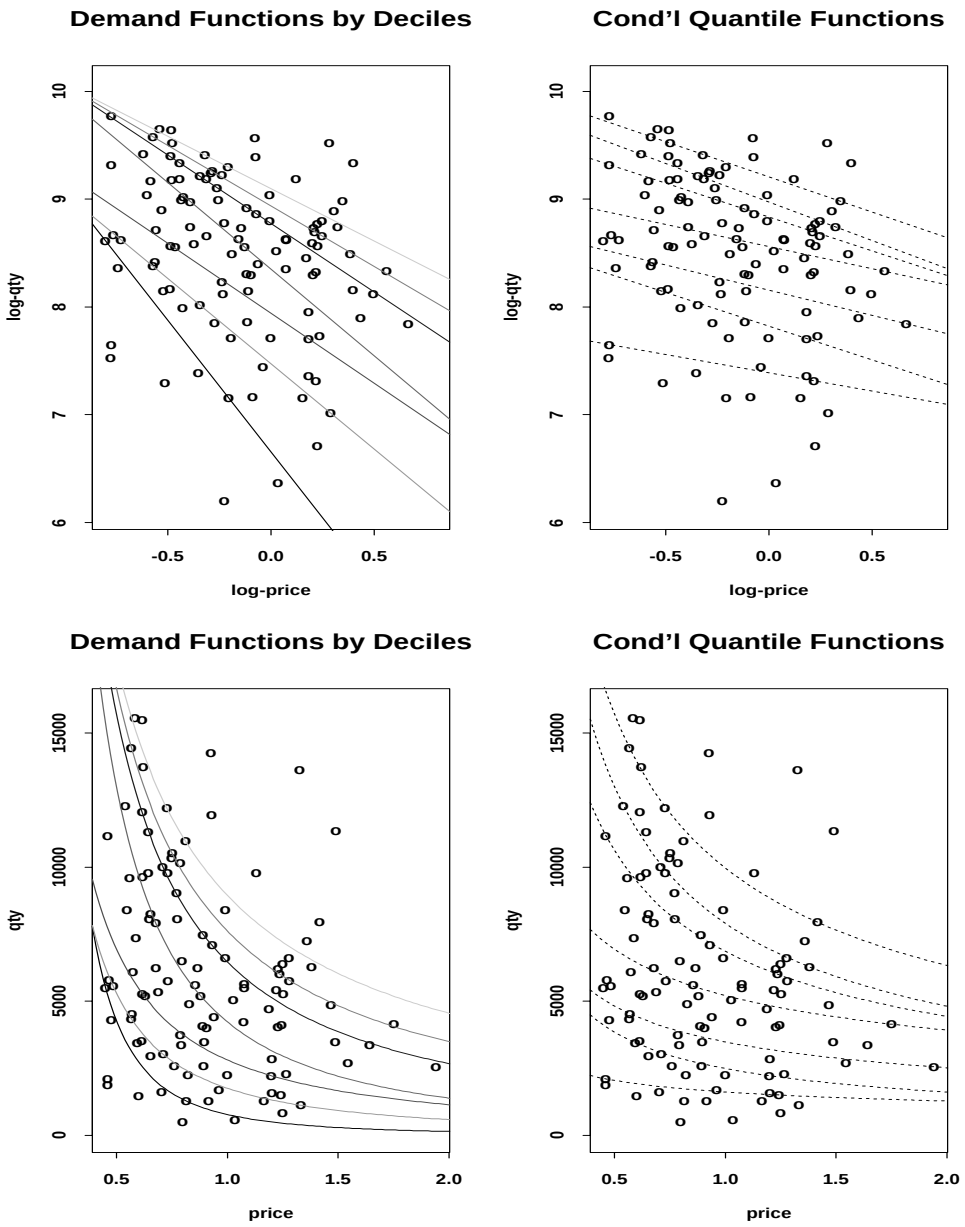

Figure 2: *Left Column*: The estimated by IQR demand curves, indexed by the quantile index (.1, .2, .3, .5, .7, .8. and .9.). The top display is in log-price-log-quantity space. The bottom display is in the original space. *Right Column*: The estimated conditional quantile curves of fish quantity sold as a function of price. The top display is in log-price-log-quantity space. The bottom display is in the original space.

for a person's actual training status. Adadie et. al. (2000), who used this data to estimate the quantile treatment effects of training on the earnings of participants using the LATE model outlined in Section 2.2, provide the detailed information regarding collection procedures, sample selection criteria, and institutional details of the JTPA along with additional facts and discussion about the JTPA training experiment. We limit the analysis to the sample of male adults.

To capture the effects of training on earnings, we estimate a linear quantile model of the form

$$Q_{Y_{d,x}}(\tau) = d\alpha(\tau) + x'\beta(\tau),$$

where d indicates training status and is instrumented for by assignment to the control group, the potential outcomes Y are earnings, and x is a vector of covariates. The data consist of 5,102 observations with data on earnings, training and assignment status, and other individual characteristics. Earnings is measured as total earnings over the 30 month period following the random assignment into the treatment or control group. We also include dummies for black and Hispanic persons, a dummy indicating high-school graduates and GED holders, five age-group dummies, a marital status dummy, a dummy indicating whether the applicant worked 12 or more weeks in the 12 months prior to the assignment, a dummy signifying that earnings data are from a second follow-up survey, and dummies for the recommended service strategy.[9]

Results for standard quantile regression are illustrated in Figure 4 and IQR estimates in Figure 3. The shaded region represents the 80 percent confidence interval for the point estimates. The first panel in each figure shows the estimated impact of the participation in the training program across various quantiles. A quick comparison of the two sets of results shows that the standard quantile regression estimates of the *statistical* impacts of training overstate the treatment effect. The quantile regression estimates are uniformly larger than the IQR estimates, and in many cases the difference is quite substantial. This difference is perhaps most important in the low to middle quantiles where the conventional quantile regression estimates indicate a positive and significant *statistical* impact of training on the earnings of participants, given that the compliance is *self-selected*.

The differences in the standard QR and the IQR estimates, as well as the distributional impacts of the program, are made even more apparent when one considers the impact of training in percentage terms.[10] Quantile regression estimates indicate large percentage statistical impacts, especially in the lower quantiles. The IQR estimates, on the other hand, indicate that the percentage *causal* impact of the training program is relatively constant and low, between 5 and 10 percent, along the whole distribution. This is interesting since the supposed intent of job training programs is to raise the in-

[9] The recommended service strategy was broken into three categories: classroom training, on-the-job training and/or job search assistance, and other forms of training.

[10] The percentage impact of training is calculated for both whites, Percentage Impact I, and black, Percentage Impact II. Percentages are calculated for married high-school graduates aged 30 to 35.

comes of low income individuals. However, we observe that the impacts were actually the greatest for the upper quantiles.

Coefficient estimates for several of the covariates are also included in the figures. None of the results are particularly surprising, and the coefficients seem to be quite well-behaved. Being Hispanic has no significant impact on potential earnings at any point in the distribution, while at medium and high quantiles, blacks earn significantly less than whites. We also see that education, as measured by high school graduation or having a GED, has a positive impact along almost the entire distribution, with the impact growing monotonically in the quantile index. This pattern is also observed for the effect of marriage, with the effect tapering off in the highest quantiles. The effect of having worked little in the previous year runs in almost exactly the opposite direction, impacting earnings negatively at all quantiles and decreasing earnings substantially in the upper tail of the distribution.

A final point is that our estimation results do not contradict those obtained by AAI. Since identification in two models comes through different assumptions *and* the estimated treatment effects are for different populations (AAI's model is for the subpopulation of compliers), it is obviously not necessary that both estimation procedures yield agreeable results. The JTPA, however, seems to be an example where *both* sets of assumptions hold. Independence and monotonicity are almost certainly satisfied, and it seems reasonable that, relative to others with similar characteristics, subjects would forecast anticipate the same ranks across treatment states, satisfying our similarity assumption[11]. Under these conditions, the models *overlap* and the results *must* indeed be similar *if* the subpopulation of compliers is *representative* of the entire population. It appears that this is the case in

7.3 Numerical Performance

The objective functions for selected quantiles from Examples 1 and 2 are graphed in Figure 5. The upper three panels in the figure illustrate the objective functions from the fish example, while the lower panels correspond to the JTPA example. The most striking feature of the pictures is how well-behaved the objective functions appear, especially in the JTPA example. Each of the objective functions from the fish example does have numerous local minima, which is likely largely attributable to the small sample size. However, in nearly all cases, the functions have an obvious unique global minimum.

[11] Note that in this case, conditioning on covariates may be important as it weakens the required similarity condition requiring that similarity only hold for people with the same covariate values.

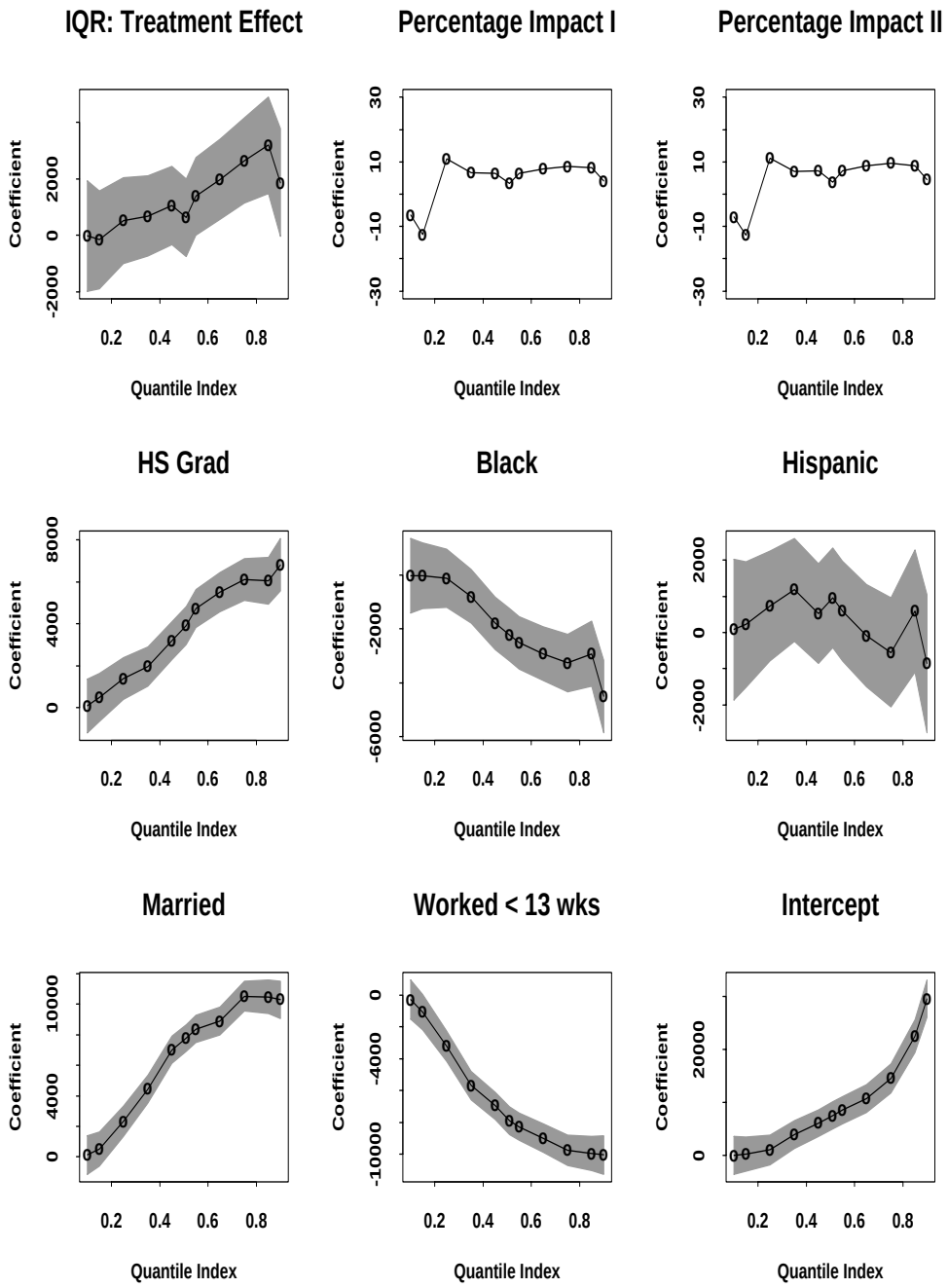

Figure 3: Inverse Quantile Regression results on JTPA data.

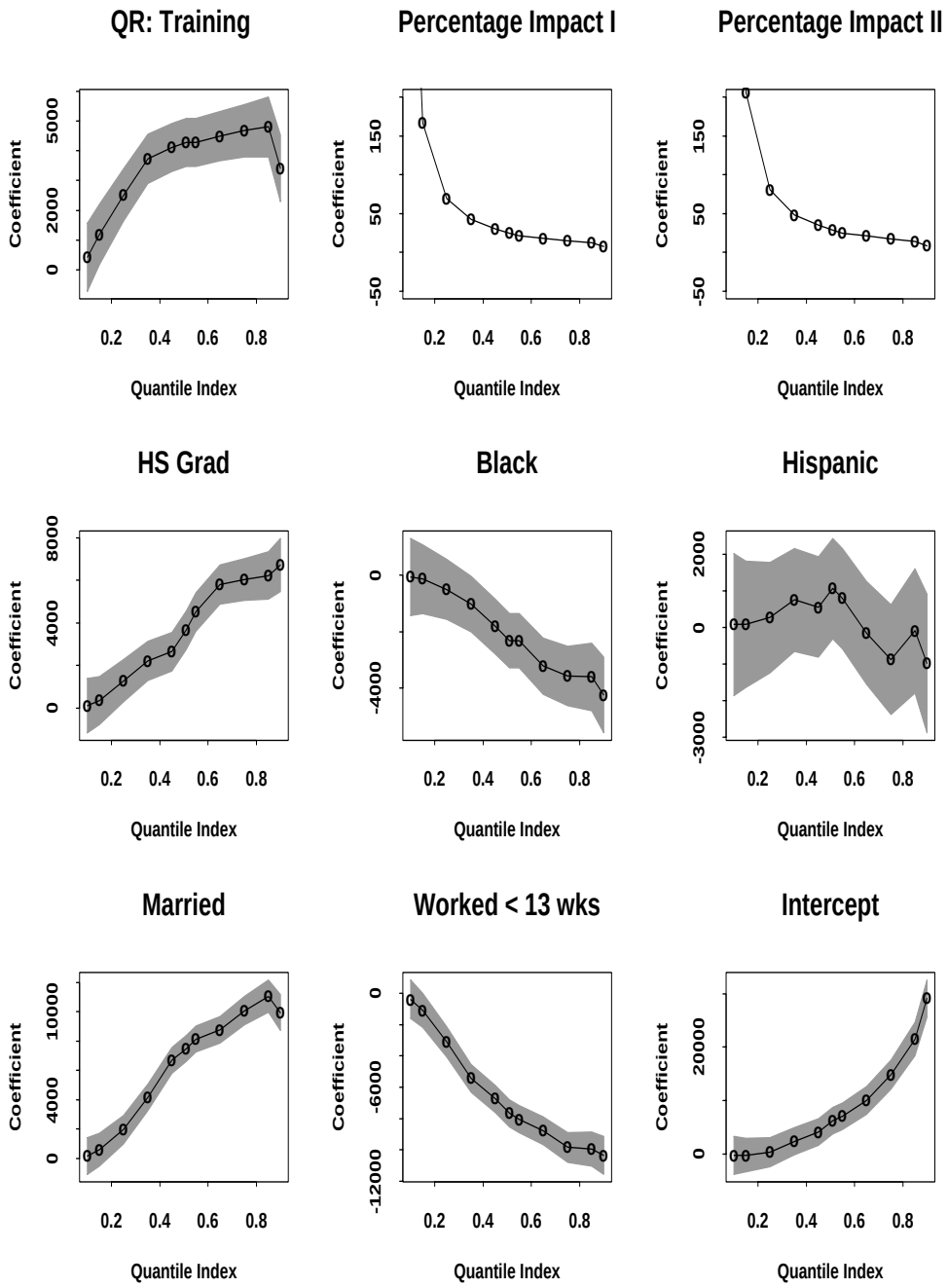

Figure 4: Quantile Regression results on JTPA data.

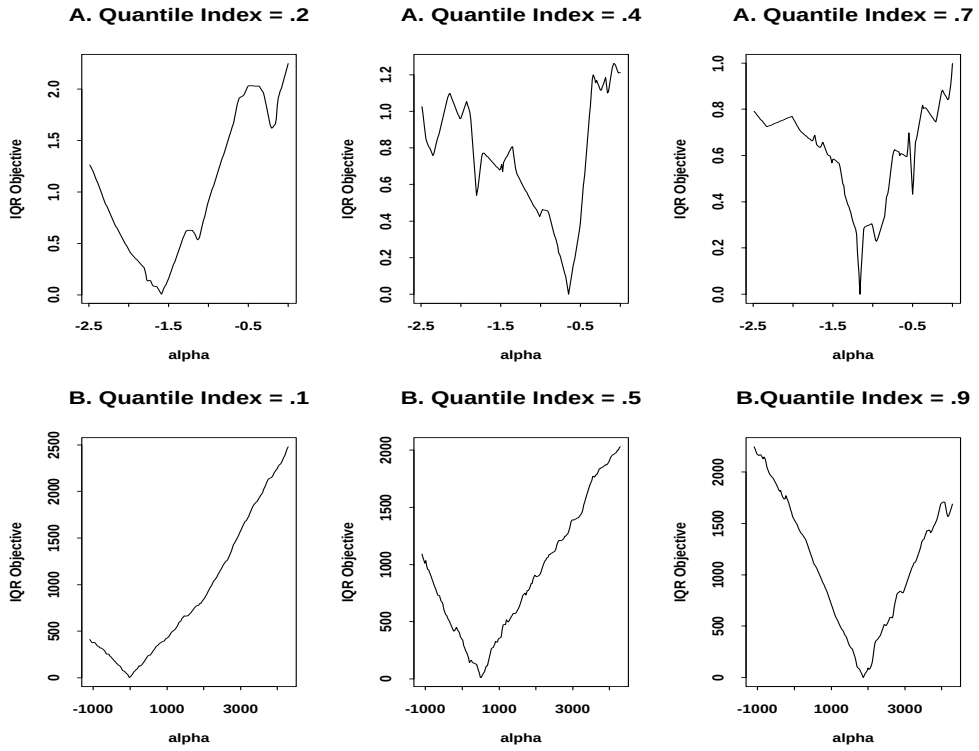

Figure 5: A. IQR objective functions for the fish example. B. IQR objective functions for the JTPA example.

8 Conclusion

The proposed QTE model and the inverse quantile regression estimators should be useful in a very wide variety of applications.

A Definitions and Lemmas

A.1 Empirical Processes

We use the following maps in the sequel

$$f \mapsto \mathbb{E}_n f(W) \equiv \frac{1}{n}\sum_{i=1}^{n} f(W_i),$$
$$f \mapsto \mathbb{G}_n f(W_i) \equiv \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \left(f(W_i) - Ef(W_i)\right).$$

Outer/inner probabilities P^*, P_* and stochastic equicontinuity etc. are as in ?. We will say that process $\{l \mapsto v_n(l), l \in \mathcal{L}\}$ is *stochastically equi-continuous* (s.e) in $\ell^\infty(\mathcal{L})$ if $\forall \epsilon > 0$ and $\eta > 0, \exists \delta > 0$:

$$\limsup_{n\to\infty} \; P^*\left(\sup_{\rho(l,l')<\delta} |v_n(l) - v_n(l')| > \eta\right) < \epsilon$$

for some semi-metric ρ on $\mathcal{L}$, such that $(\mathcal{L}, \rho)$ is totally bounded.

A.2 Argmin Convergence Lemmas

The following result is from Knight (2000). It allows for general discontinuities and $\bar{\mathbb{R}}$ -valued objective functions. Related literature is Rockafellar and Wets (1998), Salinetti and Wets (1986), Dolecki, Salinetti, and Wets (1983).

Lemma A.1 (Convexity) *Suppose $\{Q_n\}$ is a sequence of lower-semi-continuous convex $\bar{\mathbb{R}}$-valued random functions, defined on $\mathbb{R}^d$, and let $\mathcal{D}$ be a countable dense subset of $\mathbb{R}^d$. If Q_n converges to Q_∞ in $\bar{\mathbb{R}}$ on $\mathcal{D}$, in finite dimensional sense, where Q_∞ is lsc convex and finite on an open non-empty set a.s., then*

$$\underset{z\in\mathbb{R}^d}{arginf}\, Q_n(z) \xrightarrow{d} \underset{z\in\mathbb{R}^d}{arginf}\, Q_\infty(z),$$

provided the latter is uniquely defined a.s. in $\mathbb{R}^d$.

Another handy result is as follows, e.g. Knight (2000). It allows for approximate argmins and discontinuous objective functions.

Lemma A.2 (Approximate Argmins) *Suppose*

i. Z_n *is s.t.* $Q_n(Z_n) \leq \inf_{z\in\mathbb{R}^d} Q_n(z) + \epsilon_n$, $\epsilon_n \searrow 0$; $Z_n = O_p(1)$

ii. $Z_\infty \equiv argmin_{z\in\mathbb{R}^d} Q_\infty(z)$ *is uniquely defined in* $\mathbb{R}^d$ *a.s.*

iii. $Q_n(\cdot) \Rightarrow Q_\infty(\cdot)$ *in* $\ell^\infty(K)$ *over any compacts* K*, where* Q_∞ *is continuous, then* $Z_n \xrightarrow{d} Z_\infty$

B Proof of Theorem 1.

Part (a). Dependence on $X = x$ is suppressed. For P-a.e. value z of Z

$$\begin{aligned}
&P\left[Y \leq q\left[D,\tau\right]|Z=z\right] \\
&\qquad \overset{(1)}{=} P\left[q\left[D,U_D\right] \leq q\left[D,\tau\right]|Z=z\right] \\
&\qquad \overset{(2)}{=} P\left[U_D \leq \tau|Z=z\right], \\
&\qquad \overset{(3)}{=} \int P\left[U_D \leq \tau|Z=z, V_z=v\right] dP\left[V_z=v|Z=z\right] \\
&\qquad \overset{(4)}{=} \int P\left[U_{\delta(z,v)} \leq \tau|Z=z, V_z=v\right] dP\left[V_z=v|Z=z\right] \\
&\qquad \overset{(5)}{=} \int P\left[U_o \leq \tau|Z=z, V_z=v\right] dP\left[V_z=v|Z=z\right] \\
&\qquad \overset{(6)}{=} P\left[U_o \leq \tau|Z=z\right] \\
&\qquad \overset{(7)}{=} \tau.
\end{aligned} \tag{18}$$

Equality (1) is by A1 and A5. Equality (3) is a definition. (4) is by A2. (5) is by the similarity assumption A4: for each d, conditional $(V = v, X = x, Z = z)$

$$U_{\delta(z,v)} \text{ equals in distribution to } U_o.$$

(6) is a definition and (7) is by A3. Note that (2) is immediate when $\tau \mapsto q(d,\tau)$ is continuous, since we assumed that $\tau \mapsto q(d,\tau)$ is strictly increasing. To show (2) and (4) more generally, simply note that the event $\{U_D \leq \tau\}$, for $\tau \in (0,1)$, implies the event $\{q\left[D,U_D\right] \leq q\left[D,\tau\right]\}$ by $\tau \mapsto q\left[d,\tau\right]$ non-decreasing on $(0,1)$ for each d. On the other hand, the event $\{q\left[D,U_D\right] \leq q\left[D,\tau\right]\}$, for $\tau \in (0,1)$, implies the event $\{U_D \leq \tau\}$, since $\tau \mapsto q\left[d,\tau\right]$ is strictly-increasing and left-continuous in $(0,1)$ for each d.

Since $\tau \mapsto q[d,\tau]$ is strictly increasing, left-continuous, and because by steps (4)-(7) above for P-a.e. z

$$P[U_D \leq \tau|Z=z] = \tau,$$

we have $P\left[q[D,U_D] = q[D,\tau]|Z=z\right] = 0$, so that

$$1\{Y = q[D,U_D] \leq q[D,\tau]\} = 1\{Y = q[D,U_D] < q[D,\tau]\}$$

$P[\cdot|z]$-a.e., for P-a.e z. ■

Part (b). Dependence on $X = x$ is suppressed. For P-a.e. value z of Z

$$\begin{aligned}
&P\left[Y \leq q\left[D,\tau\right]|Z=z\right] \\
&\qquad \overset{(1)}{=} P\left[q\left[D,U_D\right] \leq q\left[D,\tau\right]|Z=z\right] \\
&\qquad \overset{(2)}{\geq} P\left[U_D \leq \tau|Z=z\right], \\
&\qquad \overset{(3)}{\geq} P\left[U_o \leq \tau|Z=z\right] = \tau,
\end{aligned} \tag{19}$$

where (1) and (3) are by the same arguments as in the proof of part (a), and (2) follows because the event $\{U_D \leq \tau\}$ is a subset of the event $\{q[D, U_D] \leq q[D, \tau]\}$ by $\tau \mapsto q[d, \tau]$ non-decreasing for each d. On the other hand,

$$\begin{aligned} & P\left[Y < q[D, \tau] \,|Z = z\right] \\ & \quad \overset{(4)}{=} P\left[q[D, U_D] < q[D, \tau] \,|Z = z\right] \\ & \quad \overset{(5)}{\leq} P\left[U_D < \tau | Z = z\right], \\ & \quad \overset{(6)}{\leq} P\left[U_o < \tau | Z = z\right] = \tau \end{aligned} \tag{20}$$

where (4) and (6) are by the same arguments as in the proof of part (a), and (5) follows because the event $\{q[D, U_D] < q[D, \tau]\}$ is a subset of the event $\{U_D < \tau\}$, by $\tau \mapsto q[d, \tau]$ non-decreasing and left-continuous for each d. ■

C Proof of Theorem 2

(1) ⇔ (2) follows by definition of conditional quantiles as being the left-continuous inverse of the conditional distribution function. (2) ⇔ (3) because 0 is the conditional quantile and is equivalent to the best predictor under asymmetric loss, cf. Manski (1985). Equivalence (3) ⇔ (4) is obvious. ■

D Assumptions L.1-L.6

The following assumptions are maintained for generalized empirical likelihood.

L1 $W_i = (Y_i, D_i, X_i, Z_i)$ is *i.i.d.* and (D_i, X_i, Z_i) take values in a compact set.

L2 $\vartheta \equiv (\alpha, \beta) \in \mathcal{V}$, $\mathcal{V}$ is a compact set.

L3 $\vartheta_0 \equiv (\alpha(\tau), \beta(\tau)) \in$ interior of $\mathcal{V}$, is unique ϑ s.t. $E\varphi_\tau(Y - D'\alpha - X'\beta)\Psi = 0$, in $\mathcal{V}$.

L4 $S(\vartheta) = E\varphi_\tau(Y - D'\alpha - X'\beta)^2\Psi\Psi'$ is positive definite for each $\vartheta \in \mathcal{V}$. $(S \equiv S(\vartheta_0))$

L5 $(z, x) \mapsto \hat{\Psi}(z, x) \in \mathcal{F}$, wp $\to 1$, $\mathcal{F}$ is a set of boundedly differentiable functions C^η, with smoothness order $\eta > \dim(z, x)/2$. $\hat{\Psi}(\cdot) \xrightarrow{p} \Psi(\cdot) \in \mathcal{F}$, uniformly over compacts.

L6 $J(\vartheta) \equiv \frac{\partial}{\partial v} E\left[\varphi_\tau(Y - D'\alpha - X'\beta)\Psi\right] = E\left[f_\epsilon(0|X, Z)\Psi[D' : X']\right]$ is finite, has full column rank and continuous at each ϑ in an open ball at ϑ_0. $(J \equiv J(\vartheta_0))$

Remark D.1 All assumptions, but L5, are fairly standard for EL.

Remark D.2 L5 allows for a wide variety of nonparametric and parametric estimators. For example, consistent kernel estimators with smooth kernels, various series estimators, and the smooth in (x, z) estimated parametric models for $\widehat{\Psi}(x, z)$ will satisfy this assumption. For many non-parametric examples, see Andrews (1994). In fact, L5 needs to hold only for the non-discrete sub-component of (x, z).

E Proof of Theorem 4

1. Define

$$f(W,\vartheta) \equiv \varphi_\tau(Y - D'\alpha - X'\beta)\Psi,$$
$$\hat{f}(W,\vartheta) \equiv \varphi_\tau(Y - D'\alpha - X'\beta)\widehat{\Psi},$$

where by Ψ and $\hat{\Psi}$ we denote vectors $\Psi(X,Z)$ and $\widehat{\Psi}(X,Z)$. Define also

$$Q_n(\vartheta,\gamma) \equiv \mathbb{E}_n\, \mathbf{s}[\hat{f}(W,\vartheta)'\gamma], \quad Q(\vartheta,\gamma) \equiv E\ \mathbf{s}[f(W,\vartheta)'\gamma],$$

and

$$\hat{\gamma}(\vartheta) \equiv \arg\inf_\gamma Q_n(\vartheta,\gamma), \qquad \gamma(\vartheta) \equiv \arg\inf_\gamma Q(\vartheta,\gamma),$$
$$\hat{\vartheta} \equiv \arg\sup_{\vartheta\in\mathcal{V}} \inf_\gamma Q_n(\vartheta,\gamma), \qquad \vartheta^* \equiv \arg\sup_{\vartheta\in\mathcal{V}} \inf_\gamma Q(\vartheta,\gamma).$$

By the argument of Kitamura and Stutzer (1997), $\vartheta^* = \vartheta_0$ and $\gamma(\vartheta^*) = 0$.
2. By Lemma H.1, in $\bar{\mathbb{R}}$, for any $\vartheta_n \xrightarrow{p} \vartheta_0$

$$\mathbb{E}_n\, \mathbf{s}[\hat{f}(W,\vartheta_n)'\gamma] \xrightarrow{p} E\mathbf{s}\left[f(W,\vartheta_0)'\gamma\right]$$

for each γ in a dense countable subset of $\mathbb{R}^{\dim(\gamma)}$. Hence by convexity lemma A.1, since $E\mathbf{s}\left[f(W,\vartheta_0)'\gamma\right]$ is finite over an open set by L1

$$\hat{\gamma}(\vartheta_0) \xrightarrow{p} 0, \quad \hat{\gamma}(\hat{\vartheta}) \xrightarrow{p} 0,$$

provided $\hat{\vartheta} \xrightarrow{p} \vartheta_0$.
3. By replicating the proof of Theorem 1 in Kitamura and Stutzer (1997), equations (13)-(20), replacing exp function with $\mathbf{s}$, where appropriate, we have, utilizing L5,

$$\hat{\vartheta} \xrightarrow{p} \vartheta_0.$$

4. Step 4, proved below, shows

$$\sqrt{n}\mathbb{E}_n\, \hat{f}(W,\hat{\vartheta}) = O_p(1).$$

5. In view of steps 2-3, by Lemma H.1 and properties of $\mathbf{s}$, the following expansion of the first order conditions is valid, for $(\gamma_n,\vartheta_n) = (\hat{\gamma},\hat{\vartheta})$ or $(\gamma_n,\vartheta_n) = (\hat{\gamma}(\vartheta_0),\vartheta_0)$,

$$\begin{aligned} 0 &= \sqrt{n}\mathbb{E}_n\, \hat{f}(W,\vartheta_n)\mathbf{s}[f(W,\vartheta_n)\gamma_n] \\ &= \left[\sqrt{n}\mathbb{E}_n\, \hat{f}(W,\vartheta_n)\right] + \mathbb{E}_n\, \hat{f}(W,\vartheta_n)\hat{f}(W,\vartheta_n)'\sqrt{n}\gamma_n + O_p(\sqrt{n}\|\gamma_n\|^2) \qquad (21) \\ &= \left[\sqrt{n}\mathbb{E}_n\, f(W,\vartheta_0) + (J + o_p(1))\sqrt{n}(\vartheta_n - \vartheta_0)\right] + (S + o_p(1))'\sqrt{n}\gamma_n + O_p(\sqrt{n}\|\gamma_n\|^2). \end{aligned}$$

By step 4, (21), and L4, $\sqrt{n}\gamma_n = O_p(1)$, i.e.

$$\sqrt{n}\hat{\gamma} = O_p(1) \text{ and } \sqrt{n}\hat{\gamma}(\vartheta_0) = O_p(1) \qquad (22)$$

and by (21), (22), and L6,

$$\sqrt{n}(\hat{\vartheta} - \vartheta_0) = O_p(1). \qquad (23)$$

6. Step 6, proved below, shows

$$\sqrt{n}J'\hat{\gamma} = o_p(1). \tag{24}$$

7. From (21) $\sqrt{n}\hat{\gamma} = -S^{-1}\sqrt{n}\mathbb{E}_n f(W,\vartheta_0) - S^{-1}J\sqrt{n}(\hat{\vartheta}-\vartheta_0) + o_p(1)$, which when put into (24) gives $J'S^{-1}\sqrt{n}\mathbb{E}_n f(W,\vartheta_0) + J'S^{-1}J\sqrt{n}(\hat{\vartheta}-\vartheta_0) + o_p(1) = 0$, which yields

$$\begin{aligned}
\sqrt{n}(\hat{\vartheta}-\vartheta_0) &= -(J'S^{-1}J)^{-1}J'S^{-1}\sqrt{n}\mathbb{E}_n f(W,\vartheta_0) + o_p(1) \xrightarrow{d} N(0,(J'S^{-1}J)^{-1}),\\
\sqrt{n}\hat{\gamma} &= -[S^{-1} - S^{-1}J(J'S^{-1}J)^{-1}J'S^{-1}]\sqrt{n}\mathbb{E}_n f(W,\vartheta_0) + o_p(1)\\
&\xrightarrow{d} N(0, S^{-1}[I - J(J'S^{-1}J)^{-1}J']S^{-1}),
\end{aligned}$$

and also jointly, with asymptotic covariance between $\sqrt{n}(\hat{\vartheta}-\vartheta_0)$ and $\sqrt{n}\hat{\gamma}$ equal 0. ■

Proof of step 4. By definition, for any $g_n = O_p(1/\sqrt{n})$,

$$\begin{aligned}
-n(\mathbb{E}_n \mathbf{s}[\hat{f}(W,\hat{\vartheta})'g_n] - \mathbf{s}(0)) &\le -n(\mathbb{E}_n \mathbf{s}[\hat{f}(W,\hat{\vartheta})'\hat{\gamma}] - \mathbf{s}(0))\\
&\le -n(\mathbb{E}_n \mathbf{s}[\hat{f}(W,\vartheta_0)'\hat{\gamma}(\vartheta_0)] - \mathbf{s}(0)).
\end{aligned} \tag{25}$$

By Lemma H.1, wp $\to 1$ the following expansions are valid (by steps like in (21))

$$\begin{aligned}
\text{rhs of (25)} &= \sqrt{n}\mathbb{E}_n[\hat{f}(W,\vartheta_0)]\hat{\gamma}(\vartheta_0)\sqrt{n} + \frac{1}{2}\sqrt{n}\hat{\gamma}(\vartheta_0)'S\hat{\gamma}(\vartheta_0)\sqrt{n}\\
&\quad + O_p(n\|\hat{\gamma}(\vartheta_0)\|^3) = O_p(1),
\end{aligned} \tag{26}$$

(since $\sqrt{n}\hat{\gamma}(\vartheta_0) = O_p(1)$), and

$$\text{lhs of (25)} = \sqrt{n}\mathbb{E}_n[\hat{f}(W,\hat{\vartheta})]g_n\sqrt{n} + \frac{1}{2}\sqrt{n}g_n'Sg_n\sqrt{n} + O_p(n\|g_n\|^3). \tag{27}$$

By (25)- (27), because $g_n = O_p(1/\sqrt{n})$ is arbitrary, we have $\sqrt{n}\mathbb{E}_n\hat{f}(W,\hat{\vartheta}) = O_p(1)$. ■

Proof of step 6. For any $\vartheta_n = O_p(1/\sqrt{n})$, by definition

$$-n(\mathbb{E}_n \mathbf{s}[\hat{f}(W,\hat{\vartheta})'\hat{\gamma}] - \mathbf{s}[0]) \le -n(\mathbb{E}_n \mathbf{s}[\hat{f}(W,\vartheta_n)'\hat{\gamma}] - \mathbf{s}[0]). \tag{28}$$

By Lemma H.1 and step 5, the following expansions (by steps like in (21)) are valid

$$\begin{aligned}
\text{lhs of (28)} &= -\sqrt{n}\mathbb{E}_n\hat{f}(W,\hat{\vartheta})'\hat{\gamma}\sqrt{n} - \frac{1}{2}\sqrt{n}\hat{\gamma}'S\hat{\gamma}\sqrt{n} + o_p(1),\\
\text{rhs of (28)} &= -\sqrt{n}\mathbb{E}_n\hat{f}(W,\vartheta_n)'\hat{\gamma}\sqrt{n} - \frac{1}{2}\sqrt{n}\hat{\gamma}'S\hat{\gamma}\sqrt{n} + o_p(1),
\end{aligned} \tag{29}$$

and by Lemma H.1

$$\begin{aligned}
\sqrt{n}\mathbb{E}_n\hat{f}(W,\vartheta_n) &= \sqrt{n}\mathbb{E}_n f(W,\vartheta_0) + J(\vartheta_n - \vartheta)\sqrt{n} + o_p(1),\\
\sqrt{n}\mathbb{E}_n\hat{f}(W,\hat{\vartheta}) &= \sqrt{n}\mathbb{E}_n f(W,\vartheta_0) + J'(\hat{\vartheta} - \vartheta)\sqrt{n} + o_p(1).
\end{aligned} \tag{30}$$

Putting (28) - (30) together, we have

$$\sqrt{n}(\vartheta_n - \hat{\vartheta})'J'\hat{\gamma}\sqrt{n} \le o_p(1). \tag{31}$$

Because (31) holds for any $\vartheta_n = O_p(1/\sqrt{n})$, inequality in (31) can be replaced by equality, which then implies $J'\hat{\gamma}\sqrt{n} = o_p(1)$. ■.

F Assumptions R.1-R.6

The following assumptions are maintained for the inverse quantile regression.

R1 $W_i = (Y_i, D_i, X_i, Z_i)$ is *i.i.d.* and (D_i, X_i, Z_i) take values in a compact set.

R2 $\vartheta = (\alpha, \beta) \in \mathcal{V}$, $\mathcal{V}$ is a compact set.

R3 $(\alpha_0, \beta_0) \equiv (\alpha(\tau), \beta(\tau)) \in \text{ int}\mathcal{V}$ is unique ϑ s.t. $E\varphi_\tau(Y - D'\alpha - X'\beta)\Psi \cdot V = 0$, in $\mathcal{V}$.

R4 Y has P-a.e. bounded conditional density given X, Z.

R5 $(z, x) \mapsto \hat{\Psi}(z, x)$ and $(z, x) \mapsto \hat{V}(z, x) \in \mathcal{F}$, wp $\to 1$, $\mathcal{F}$ is a set of boundedly differentiable functions C^η, with smoothness order $\eta > \dim(z, x)/2$. $\hat{\Phi}(\cdot) \xrightarrow{p} \Phi(\cdot), \hat{V}(\cdot) \xrightarrow{p} V(\cdot) \in \mathcal{F}$, uniformly over compact sets. $V(\cdot) > \delta > 0$.

R6 $J(\pi) \equiv \frac{\partial}{\partial \pi} E\left[\varphi_\tau(Y - D'\alpha - X'\beta - \Phi' Z)\Psi\right]$ has full column rank and is continuous at each $\pi = (\alpha, \beta, \gamma)$ in an open ball at $(\alpha(\tau), \beta(\tau), 0)$

Remark F.1 All assumptions, but R5, are analogs of standard conditions for quantile regression. These could be refined at a notational cost.

Remark F.2 R5 allows for a wide variety of nonparametric and parametric estimators. Recall that $\hat{\Phi}$ is the estimated instrument $\hat{\Phi}(X, Z)$ and $\hat{V} \equiv \hat{V}(X, Z)$ is the estimated weight in the formulation of IQR. For example, consistent kernel estimators with smooth kernels, various series estimators, and the smooth in (x, z) estimated parametric models for $\widehat{\Psi}(x, z)$ or $\hat{V}(x, z)$ will satisfy this assumption. For many non-parametric examples, see Andrews (1994). R5 needs to hold only for the non-discrete sub-component of (x, z).

G Proof of Theorem 5

1. Define for $\theta \equiv (\beta, \gamma)$ and $\theta_0 \equiv (\beta(\tau), 0)$

$$\hat{f}(W, \alpha, \theta) \equiv \varphi_\tau(Y - D'\alpha - X'\beta - \hat{\Phi}'\gamma)\hat{\Psi},$$
$$f(W, \alpha, \theta) \equiv \varphi_\tau(Y - D'\alpha - X'\beta - \Phi'\gamma)\Psi,$$

where $\Psi \equiv V \cdot (X',\ \Phi(X, Z)')', \Phi \equiv \Phi(X, Z)$, $\widehat{\Psi} \equiv \widehat{V} \cdot (X', \widehat{\Phi}(X, Z)')', \widehat{\Phi} \equiv \widehat{\Phi}(X, Z)$. Define

$$\hat{g}(W, \alpha, \theta) \equiv \rho_\tau(Y - D'\alpha - X'\beta - \hat{\Phi}'\gamma)\hat{V},$$
$$g(W, \alpha, \theta) \equiv \rho_\tau(Y - D'\alpha - X'\beta - \Phi'\gamma)V,$$

let $Q_n(\alpha, \theta) \equiv \mathbb{E}_n\, \hat{g}(W, \alpha, \theta)$, $\quad Q(\alpha, \theta) \equiv Eg(W, \alpha, \theta)$, and

$$\hat{\theta}(\alpha) \equiv \arg\inf_\theta Q_n(\alpha, \theta), \quad \theta(\alpha) \equiv \arg\inf_\theta Q(\alpha, \theta),$$
$$\hat{\alpha} \equiv \arg\inf_{\alpha \in \mathcal{V}} \|\hat{\gamma}(\alpha)\|, \qquad \alpha^* \equiv \arg\inf_{\alpha \in \mathcal{V}} \|\gamma(\alpha)\|.$$

By R3 and R4 $\alpha^* = \alpha_0$, $\theta(\alpha^*) = \theta_0$, i.e. $\gamma(\alpha^*) = 0$, $\beta(\alpha^*) = \beta$.

2. For each α and θ, by lemma H.1

$$\mathbb{E}_n\left[\hat{g}(W, \alpha, \theta) - \hat{g}(W, \alpha_0, \bar{\theta})\right] \xrightarrow{p} E[g(W, \alpha_0, \theta) - g(W, \alpha_0, \bar{\theta})],$$

for some fixed $\bar{\theta}$ (the subtraction of the terms is to make the summands bounded). The lhs and rhs are finite convex functions in θ, α. Therefore the convergence is uniform over compacts, so in view of convexity lemma A.1

$$\hat{\theta}(\alpha_0) \xrightarrow{p} \theta_0, \ \hat{\theta}(\hat{\alpha}) \xrightarrow{p} \theta_0,$$

provided $\hat{\alpha} \xrightarrow{p} \alpha_0$.

3. By an argument like in the proof of Theorem 1 in Kitamura and Stutzer (1997), we can show $\hat{\alpha} \xrightarrow{p} \alpha_0$.

4. By computational properties of quantile regression, for any α_n in a small ball at α_0

$$O(1/\sqrt{n}) = \sqrt{n}\mathbb{E}_n \hat{f}(W, \alpha_n, \hat{\theta}(\alpha_n)).$$

By lemma H.1, the following expansion of r.h.s. is valid for any $\alpha_n \xrightarrow{p} \alpha_0$:

$$\begin{aligned}\sqrt{n}\mathbb{E}_n \hat{f}(W, \alpha_n, \hat{\theta}(\alpha_n)) &\equiv \mathbb{G}_n \hat{f}(W, \alpha_n, \hat{\theta}(\alpha_n)) + \sqrt{n}E\hat{f}(W, \alpha_n, \hat{\theta}_n(\alpha_n)) \\ &= \mathbb{G}_n f(W, \alpha_0, \theta_0) + o_p(1) + \sqrt{n}E\hat{f}(W, \alpha_n, \hat{\theta}(\alpha_n))\end{aligned}$$

Expanding the last line further

$$\begin{aligned}&= \mathbb{G}_n f(W, \alpha_0, \theta_0) + o_p(1) \\ &+ (J_\theta + o_p(1))\sqrt{n}(\hat{\theta}(\alpha_n) - \theta_0) \\ &+ (J_\alpha + o_p(1))\sqrt{n}(\alpha_n - \alpha_0)\end{aligned}$$

In other words

$$\sqrt{n}(\hat{\theta}(\alpha_n) - \theta_0) = -J_\theta^{-1}\mathbb{G}_n f(W, \alpha_0, \theta_0) - J_\theta^{-1}J_\alpha[1 + o_p(1)]\sqrt{n}(\alpha_n - \alpha_0) + o_p(1),$$

i.e.

$$\sqrt{n}(\hat{\gamma}(\alpha_n) - 0) = -\bar{J}_\gamma\mathbb{G}_n f(W, \alpha_0, \theta_0) - \bar{J}_\gamma J_\alpha[1 + o_p(1)]\sqrt{n}(\alpha_n - \alpha_0) + o_p(1),$$

Over a shrinking ball $B(\alpha_0, \delta_n)$, wp $\to 1$,

$$\hat{\alpha} = \arg \inf_{\alpha_n \in B(\alpha_0, \delta_n)} \|\hat{\gamma}(\alpha_n)\|$$

Observe

$$\sqrt{n}\|\hat{\gamma}(\alpha_n)\| = \|O_p(1) - \bar{J}_\gamma J_\alpha[1 + o_p(1)]\sqrt{n}(\alpha_n - \alpha_0)\|$$

Since $\bar{J}_\gamma J_\alpha$ has the full column rank, $\sqrt{n}(\hat{\alpha} - \alpha_0) = O_p(1)$. Hence by lemma A.2

$$\sqrt{n}(\hat{\alpha} - \alpha_0) - \arg\inf_\mu \| - \bar{J}_\gamma\mathbb{G}_n f(W, \alpha_0, \theta_0) - \bar{J}_\gamma J_\alpha \mu\| \xrightarrow{p} 0.$$

So the conclusion now follows:

$$\sqrt{n}(\hat{\alpha} - \alpha_0) = -(J_\alpha'\bar{J}_\gamma'\bar{J}_\gamma J_\alpha)^{-1}J_\alpha'\bar{J}_\gamma'\bar{J}_\gamma\mathbb{G}_n f(W, \alpha_0, \theta_0) + o_p(1)$$

and

$$\sqrt{(}\hat{\theta} - \theta_0) = -J_\theta^{-1}[I - J_\alpha(J_\alpha'\bar{J}_\gamma'\bar{J}_\gamma J_\alpha)^{-1}J_\alpha'\bar{J}_\gamma'\bar{J}_\gamma]\mathbb{G}_n f(W, \alpha_0, \theta_0) + o_p(1)$$

with $\mathbb{G}_n f(W, \alpha_0, \theta_0) \xrightarrow{d} N(0, S)$ by CLT. ■.

H A Lemma

Lemma H.1 (Expansions) *Under assumptions L1-L6, as $\vartheta \xrightarrow{p} \vartheta_0$, for any real-valued function m that is Liphitz over a compact set,*

i. $\sqrt{n}\mathbb{E}_n \hat{f}(W, \vartheta_n) = \sqrt{n}E_n f(W, \vartheta_0) + (J + o_p(1))\sqrt{n}(\vartheta_n - \vartheta_0)$,

ii. $\mathbb{E}_n m[\hat{f}(W, \vartheta_n)] \xrightarrow{p} Em[f(W, \vartheta_0)]$ *(in particular, for $m(x) = xx'$),*

iii. $E_n \mathbf{s}[\hat{f}(W, \vartheta_n)'\gamma] \xrightarrow{p} E\mathbf{s}[f(W, \vartheta_0)'\gamma]$ *for each γ in a countable dense subset of $\mathbb{R}^{\dim(\gamma)}$.*

Under assumption R1-R6,

iii. For each α and θ,

$$\mathbb{E}_n [\hat{g}(W, \alpha, \theta) - \hat{g}(W, \alpha_0, \bar{\theta})] \xrightarrow{p} E[g(W, \alpha_0, \theta) - g(W, \alpha_0, \bar{\theta})],$$

iv. $\mathbb{G}_n \hat{f}(W, \alpha_n, \hat{\theta}(\alpha_n)) = \mathbb{G}_n f(W, \alpha_0, \theta_0) + o_p(1)$, *for any* $\alpha_n \xrightarrow{p} \alpha_0$.

Proof. i. The class of functions

$$\mathcal{H} \equiv \left\{(\Psi, \vartheta) \mapsto \varphi_\tau(Y - D'\alpha - X'\beta)\Psi(x, z), \quad \vartheta \in B(\vartheta_0, \epsilon), \Psi \in \mathcal{F}\right\}$$

is Donsker. The bracketing number of $\mathcal{F}$ satisfies by Cor 2.7.4 in van der Vaart and Wellner (1996)

$$\log N_{[\cdot]}(\epsilon, \mathcal{F}, L_2(P)) = O\left(\frac{1}{\epsilon}^{\frac{\dim(z,x)}{\eta}}\right) = O\left(\frac{1}{\epsilon}^{2+\delta'}\right),$$

for some $\delta' > 0$. Thus $\mathcal{F}$ is Donsker. Exploiting L6 and monotonicity of the indicator, the bracketing number of

$$\mathcal{G} \equiv \left\{\vartheta \mapsto \varphi_\tau(Y - D'\alpha - X'\beta), \quad \vartheta \in B(\vartheta_0, \epsilon)\right\}$$

satisfies

$$\log N_{[\cdot]}(\epsilon, \mathcal{G}, L_2(P)) = O\left(\ln \frac{1}{\epsilon}^{\dim(\alpha)+\dim(\beta)}\right),$$

so $\mathcal{G}$ is Donsker as well. Class $\mathcal{H}$ is formed as a product of these two uniformly bounded (by L1 and L5) classes:

$$\mathcal{H} = \mathcal{F}\mathcal{G},$$

so the product is Lipshitz over $(\mathcal{F} \times \mathcal{G})$, and by Theorem 2.10.6 in van der Vaart and Wellner (1996) $\mathcal{H}$ is Donsker.

Define the process

$$h = (\Psi, \vartheta) \mapsto \mathbb{G}_n \varphi_\tau(Y - D'\alpha - X'\beta)\Psi(X, Z)$$

Since $\hat{\Psi} \xrightarrow{p} \Psi_0$ uniformly over compacts and $\vartheta \xrightarrow{p} \vartheta_0$, we have $\rho(\hat{h}, h) \xrightarrow{p} 0$, where ρ is defined by the $L_2(P)$ seminorm $\rho(h) \equiv E\|\varphi_\tau(Y - D'\alpha - X'\beta)\Psi(X, Z) - E\varphi_\tau(Y - D'\alpha - X'\beta)\Psi(X, Z)\|$, so that

$$\mathbb{G}_n \varphi_\tau(Y - D'\alpha_n - X'\beta_n)\hat{\Psi}(X, Z) - \mathbb{G}_n \varphi_\tau(Y - D'\alpha - X'\beta)\Psi(X, Z) = o_p(1)$$

Next note

$$\sqrt{n}\mathbb{E}_n \hat{f}(W, \vartheta_n) = \mathbb{G}_n \hat{f}(W, \vartheta_n) + E\hat{f}(W, \vartheta_n) = \mathbb{G}_n f(W, \vartheta_0) + o_p(1) + \sqrt{n}E\hat{f}(W, \vartheta_n),$$

and **i.** follows by Taylor expansion of $E\hat{f}(W,\vartheta_n)$ in ϑ_n: $E\hat{f}(W,\vartheta_n) = (J+o_p(1))\sqrt{n}(\vartheta_n - \vartheta_0)$.

By the above analysis $G_n m[\hat{f}(W,\vartheta_n)]$ is Donsker (asymptotically Gaussian) as well, using Theorem 2.10.6 in van der Vaart and Wellner (1996) (m is Liphitz over bounded subsets to which $\hat{f}(W,\vartheta_n)$ belongs wp $\to$ 1 by assumption.) From this ii. is immediate.

To show iii. note that γ is either such that $E\mathbf{s}[f(W,\vartheta_0)'\gamma] = +\infty$ or $E\mathbf{s}[f(W,\vartheta_0)'\gamma] < \infty$. By convexity and lower-semicontinuity of $\mathbf{s}$, the latter set, say F, is convex, open, and its boundary is nowhere dense in $\mathbb{R}^{\dim(\gamma)}$. Thus for $\gamma \in F$, $E\mathbf{s}[f(W,\vartheta)'\gamma]|_{v=v_n, f=\hat{f}} < \infty$, wp $\to 1$ by continuity of this expectation implied by L6. Conditional on this event, step ii. gives $E_n\mathbf{s}[\hat{f}(W,\vartheta_n)'\gamma] \xrightarrow{p} E\mathbf{s}[f(W,\vartheta_0)'\gamma] < \infty$. Similarly take γ in $\bar{F}^c$, where $\bar{F}$ denotes the closure of F. The analogous argument delivers, $E_n\mathbf{s}[\hat{f}(W,\vartheta_n)'\gamma] \xrightarrow{p} E\mathbf{s}[f(W,\vartheta_0)'\gamma] = \infty$. So iii. follows by taking all the rationals not in the boundary of F.

Finally, v. and iv. can be proved along the lines of i. and ii., respectively. ■

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